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Computation of Multi-Particle Phase-Space Integrals and Their Applications in High-Energy Physics

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Resumo

Breve resumo em Português.

Abstract

Short abstract in English.

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Introduction

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Theoretical Foundations

The study of quarks and gluons, and related structures like those described throughout the present text, is conducted within the framework of Quantum Chromodynamics (QCD). QCD is the theory behind the Strong Interaction, one of the four fundamental interactions. QCD is a $SU(3)$ gauge theory. Its gauge boson is the gluon, which carries gauge charge, in this case, colour charge. This non-Abelian[1] nature leads to gluon self-interaction, which fundamentally changes the dynamics of the theory relative to other Abelian theories like Quantum Electrodynamics (QED). Due to this gluon self-interaction, QCD exhibits two energy-dependent behaviours: at low energy (long distances) it exhibits Colour Confinement, which binds quarks and gluons together, confining them into hadrons; at high energies (short distances), it exhibits Asymptotic Freedom, where the coupling becomes relatively small ($\alpha_s \ll 1$), allowing for quarks and gluons to behave almost like free particles.

To use QCD theory applied to hadron colliders, we rely on collinear factorisation. The problem is split into a non-perturbative long-distance part, modelled using measurable Parton Distribution Functions (PDFs), and a short-distance high-energy part, using calculable hard scattering matrix elements. Compared to collider energies q, the strong coupling constant $\alpha_s(q^2)$ tends towards zero. This allows us to calculate the hard scattering process using a perturbative Taylor-like expansion in α_s, as follows,

$$\sigma = \sigma_{\text{LO}} + \alpha_s \sigma_{\text{NLO}} + \alpha_s^2 \sigma_{\text{NNLO}} + O\left(\alpha_s^3\right). \tag{2.1}$$

The series must be truncated at a fixed order in practice. This truncation introduces artificial dependences on unphysical scales like the renormalisation (μ_R) and the factorisation scales (μ_F), respectively from regularising ultraviolet (UV) divergences and marking the boundary between the two energy regimes [1, 2, 3].

In recent years, with the opening and operation of the Large Hadron Collider (LHC) and the very large amounts of data it produces, experimental statistical uncertainties are shrinking to the 1 to 2% level, with systemic uncertainties also highly constrained. At the same time, theoretical Next-to-Leading-Order (NLO) predictions carry scale uncertainties of 10 to 15%. This fact makes the development of Next-to-Next-to-Leading-Order (NNLO) predictions necessary not only to match the most recent experimental uncertainties but also to be able to correctly predict and measure low-level deviations to accepted theory, which may open the doors to new physics. In even more recent years, the upcoming opening of the Future Circular Collider - electron-positron (FCC-ee) is predicted to lower experimental uncertainties to sub-1% level, making the development of NNLO (and even N^{3}LO) predictions increasingly essential [4, 5].

[1] A gauge theory is said to be non-Abelian when the generators of its underlying symmetry group, $SU(3)$, in this case, do not commute. This property requires the force-carrying bosons to carry the charge of the interaction themselves, leading to their self-interaction.

2.1 Divergences in Perturbative QCD

2.1.1 Infrared Divergences

The two QCD infrared limits, soft and collinear, can be derived explicitly as follows. We start by defining the physical system under study: consider two massless final-state partons i and j with four-momenta p_i and p_j. By momentum conservation, the four-momentum of the internal virtual propagator is $P = p_i + p_j$. The matrix element contains this propagator and in the square amplitude it appears as a $1/P^2$ factor. Expanding it and applying the massless condition ($p^2 = 0$), we get $P^2 = 2p_i \cdot p_j = 2E_iE_j(1 - \cos\theta_{ij})$ [1]. Importantly, this quantity defines the Mandelstam invariants as $s_{ij} = 2p_i \cdot p_j = P^2$, in the massless regime[2], which will serve as the natural integration variable throughout the work. The limits are then apparent: the soft limit is hit when one of the energies goes to zero and the collinear limit, when θ_{ij} does. In both cases, $1/s_{ij}^2 \to \infty$,

$$\begin{aligned} \lim_{E_i \to 0} s_{ij} &= 0 \\ \lim_{\theta_{ij} \to 0} s_{ij} &= 0. \end{aligned} \tag{2.2}$$

Note here that in terms of detection, a real emission that is either soft or collinear is experimentally indistinguishable from the corresponding n-particle state. This means that at the limits, there is nothing distinguishing the unresolved real emission $(n + 1)$ and the Born-level (n) cases.

Since the soft and collinear limits from real radiation are indistinguishable from the same limits in the virtual regime, cross-sections for individual degenerate states are not separately physically meaningful. The KLN theorem states that to compute a physical observable the probabilities of all degenerate states must be summed over [6, 7]. Using this theorem, we can sum all real radiation states' singularities as positive divergences and all virtual corrections' singularities as negative divergences, such that they cancel. For this cancellation to happen, however, the two parts must live within the same phase-space, which is not the case here, for these real emissions and virtual corrections, since one produces $(n + 1)$ particles and the other only n. The solution here is to subtract the divergences using a local counterterm σ^S and integrate over the $n + 1$ phase-space, and then add the subtracted divergences to the n-particle phase-space integrated analytically over the extra particle and then integrate the whole term over the phase-space. This is expressed schematically as,

$$\sigma = \int_{n+1} \left(d\sigma^R - d\sigma^S \right) + \int_n \left(d\sigma^V + \int_1 d\sigma^S \right) \tag{2.3}$$

where $d\sigma^R$ represents the differential real radiation cross-section, $d\sigma^V$ the differential virtual corrections cross-section and $d\sigma^S$ the differential subtraction counterterm [8]. Importantly, $d\sigma^S$ must reproduce $d\sigma^R$'s exact singular behaviour at every point in the phase space, such that the divergences cancel locally.

2.1.2 Ultraviolet Divergences

Infrared divergences are not the only kind of divergences we find in QCD. When we study one-loop diagrams new limits must be considered for the loop momentum l. The soft and collinear limits on l behave exactly as described above for the outgoing momenta. That is the reason $d\sigma^V$ is singular. But,

[2] The general definition of the Mandelstam invariant is $s_{ij} = (p_i + p_j)^2$, but as p_k^2 for any k, in this massless regime, then the massless Mandelstam invariant is defined as $s_{ij} = 2p_i \cdot p_j$.

since this momentum is not bound by the phase-space to a certain maximum, being instead integrated over the whole of momentum space, a new $l \to \infty$, so-called ultraviolet (UV) limit appears.

This limit presents two distinct but pressing problems. Firstly, the unbounded domain risks the integral diverging. This is due to, in d-dimensions, the loop measure expanding to,

$$d^d l = l^{d-1} dl \, d\Omega_{d-1}. \tag{2.4}$$

Whether the integral diverges or not depends on whether the integrand converges fast enough to counter the l^{d-1} measure term. Since each propagator contributes $\sim 1/l^2$ to the integrand, schematically, a diagram with n_p propagators in the loop and no extra powers of l in the numerator becomes,

$$\int^{\infty} l^{d-1} \cdot \frac{1}{l^{2n_p}} = \int^{\infty} l^{d-1-2n_p} dl \tag{2.5}$$

which is UV divergent whenever $d - 1 - 2n_p \geq -1 \Rightarrow d \geq 2n_p$ [2, 3].

Secondly, this limit tells us that any perturbative series constructed order by order in the bare coupling α_0, from the QCD Lagrangian [ref to the qcd lagrangian], is, beyond tree level ill-defined since its higher-order coefficients diverge [1]. This property makes the variable unsuitable as the order by order perturbative series variable we require. As such, we need to trade it for a finite renormalised coupling, α_s, for the order by order perturbative series [3].

2.2 Dimensional Regularization

The subtraction scheme from Eq. 2.3 cancels the real radiation poles directly using $d\sigma^S$, but the system as shown in the equation still is not numerically integrable and will still go to infinity when integrated due to the rightmost integral over the unresolved parton,

$$\int_1 d\sigma^S. \tag{2.6}$$

Since $d\sigma^S$ is designed explicitly to carry over all of $d\sigma^R$'s divergent structures, this integral blows up once numerically integrated over 4 dimensions. This is also true of the virtual corrections in $d\sigma^V$, as explored in Subsection 2.1.2. To solve this we integrate this term over the unresolved parton over d-dimensions and then use Dimensional Regularisation to manifest the physical divergences as explicit analytic poles in an arbitrary continuous parameter ϵ, via,

$$d = 4 - 2\epsilon \tag{2.7}$$

which leaves us with two ϵ-dependent expressions for the virtual corrections and for the now unresolved-parton-integrated subtraction cross-section,

$$\int_n \left(d\sigma^V + d\sigma_1^S \right). \tag{2.8}$$

From the KLN theorem, as described in Section 2.1.1, we know that these poles must cancel. This makes it so that the complete expression becomes finite and numerically integrable.

Though the principle is always the same, there is more than one way to implement Dimensional Regularisation. The most common methods are 't Hooft-Veltman [9], Four-Dimensional Helicity [10] and Conventional Dimensional Regularisation (CDR) [1]. This third method is the one used throughout the present work, since AndCalc's primary objective is the simplification and democratisation of the antenna framework establish in [11], which uses CDR throughout.

The use of CDR, specially the complete shift to d-dimensional space introduces some complications, however. First, the move to d-dimensional spherical coordinates implies that the phase-space integrals are now over d-dimensional solid angles, which produce known geometric artifacts like π and $(4\pi)^{-\epsilon}$ factors and Gamma functions like $\Gamma(1-\epsilon)$. To prevent these artefacts from contaminating our integrated results conform with the $\overline{\text{MS}}$ (modified minimal subtraction) renormalisation scheme [12], we must extract the $S_\epsilon = (4\pi)^{-\epsilon} e^{\epsilon \gamma_E}$ normalisation factor from the integration measure. In the standard QCD convention, the perturbative expansions are written in terms of $\alpha_s/(2\pi)$, instead of just α_s. That makes the introduction of a further normalisation parameter necessary, since the squared matrix elements do not scale just with the coupling constant α_s, but rather with the whole raw coupling constant $g^2 = 4\pi\alpha_s = 8\pi^2(\alpha_s/(2\pi))$. The scale normalisation,

$$\begin{aligned} G_k &= \left(8\pi^2\right)^k \\ &= \left(8\pi^2\right)^{n-2} \Lambda_l, \quad \Lambda_l \equiv \left(8\pi^2\right)^l \end{aligned} \tag{2.9}$$

is then introduced to normalise those expansion terms, and the loop-order-only normalisation Λ_l explicitly shown as the virtual-loop contribution. The combined normalisation here introduced is thus,

$$C(\epsilon, k) = G_k S_\epsilon = \left(8\pi^2\right)^k (4\pi)^{-\epsilon} e^{\epsilon \gamma_E} \tag{2.10}$$

where k represents the perturbative order as $k = 1$ for NLO, $k = 2$ for NNLO, ... and is, itself defined as,

$$k = (n-2) + l \tag{2.11}$$

for some antenna with n-final-state particles and l loops[3].

Since multi-particle phase spaces recursively factorise, we must also normalise our phase-space integral by the d-dimensional two-body phase space volume Φ_2 [4]. This normalisation, together with the previous combined normalisation factor make the global normalisation factor $\mathcal{N}(\epsilon, k)$, used throughout the present work,

$$\mathcal{N}(\epsilon, k) = \frac{C(\epsilon, k)}{\Phi_2}. \tag{2.12}$$

As described in Subsection 2.1.2, we need to trade the bare coupling constant α_0 for a suitable order by order perturbative series variable. That is achieved as follows [11],

$$\alpha_0 \mu_0^{2\epsilon} S_\epsilon = \alpha_s \mu^{2\epsilon} \left[1 - \frac{\beta_0}{\epsilon}\frac{\alpha_s}{2\pi} + O(\alpha_s^2)\right], \quad \beta_0 = \frac{11N - 2N_f}{6} \tag{2.13}$$

where μ_0 is the bare mass parameter that keeps α_0 dimensionless, μ is the renormalisation scale (the renormalised mass parameter) that functions equally for α_s. It is more commonly scaled as $\mu^2 = q^2$ for the massless case. β_0 is the one-loop coefficient of the QCD beta function, that tracks how the renormalised coupling α_s changes with the renormalisation scale μ, and finally N is the number of colours, 3, and N_f the number of massless active quark flavours.

[3]Loops will be discussed and the k index widely used in Subsection 2.3.1.

[4]This factorisation and subsequent normalisation is explored in detail in Subsection 2.3.4

2.3 The Antenna Subtraction Formalism

In perturbative QCD, when a particle like a gluon, j, becomes unresolved as described above, its squared matrix element $|\mathcal{M}_{n+1}|^2$ factorises into a universal singular factor for this process, $\mathcal{S}_j$ and a lower-multiplicity squared matrix element (a level below), as follows [8, 11],

$$|\mathcal{M}_{n+1}(...,i,j,k,...)|^2 \to \mathcal{S}_j \cdot |\mathcal{M}_n(...,I,K,...)|^2. \tag{2.14}$$

Since this is a universal property, it is possible to compute the subtraction cross-section $d\sigma^S$ by summing over all possible unresolved pairs. Therefore,

$$d\sigma^S = \sum_{\text{unresolved pairs}} \mathcal{S}_j \cdot d\sigma^B \tag{2.15}$$

where $d\sigma^B$ denotes the Born process $\gamma^* \to q\bar{q}$, cross-section and $\mathcal{S}_j$ is the singular factor for each of the j particles being summed over.

2.3.1 Antenna Functions

Since the singular factor $\mathcal{S}_j$ must work smoothly across all unresolved limits and must also handle all spin correlations correctly, it should, optimally, be a function that preserves all the unresolved parton information without the noise coming from the hard parton process. This is easily achieved by dividing our n-level process squared matrix element by the fixed two-parton Born-level, common to all quark-antiquark antenna functions. The resulting quantity is gauge-invariant. It is the antenna, most usually notated by X_n^0 and defined as [11, 13, 14],

$$X_n^{0,\,\text{bare}} = \mathcal{N}_X \frac{\left|\mathcal{M}_n^0\right|^2}{\left|\mathcal{M}_2^0\right|^2} \tag{2.16}$$

where $|\mathcal{M}_n|^2$ represents the n-level squared matrix element and $X_n^{0,\,\text{bare}}$ is the antenna that represents that same n-level. $\left|\mathcal{M}_2^0\right|^2$ is the Born-level squared matrix element. $\mathcal{N}_X$ represents the necessary normalisation over colour and coupling terms, whose general form depends on the specific antenna family under consideration. The colour-algebraic machinery required to construct it explicitly is developed in Subsection 2.3.2. It is defined, for quark-antiquark antennae [5], using the fundamental Casimir invariant C_F and the strong coupling constant $g_s = \sqrt{4\pi\alpha_s}$, as

$$\mathcal{N}_X = \frac{1}{2g_s^{2(n-2)}C_F}. \tag{2.17}$$

This definition is strictly correct and complete for real-emission, so called tree-level, antenna functions. These antenna functions represent loopless processes. QCD allows for the exchange of gluons in between quarks and the self-absorption (by the same quark) of a previously emitted gluon. These are the so-called virtual corrections. As described in Section 2.1.2, these corrections lead to UV divergences. Hence, loop antenna functions require normalisation and hence why the Eq. 2.16 is labelled as *bare*: it is the general definition without normalisation. This normalisation is achieved through the rescaling of the bare coupling constant α_0 to the renormalised coupling constant α_s through Eq. 2.13. This rescaling

[5] Other antenna types are described in Subsection 2.3.3. These require different normalisations.

applies the Ward-Takahashi [15, 16] and Slavnov-Taylor identities[6] [17, 18], which guarantee the conservation of the electric and strong charges[7], respectively, through all topologies of the same process. They also require us to denote the number of loops their process supports. That number is the superscript in X_n^l, l, which up until now had been kept at zero for the loopless processes. n, on the other hand, represents the number of final-state particles in the process.

These antenna functions may now be ordered by perturbative order based on their k index. Let us define an antenna completely as $X_{n,(k)}^l$ where k is the perturbative order index, n is the number of final-state particles and l is the number of loops. Let us also define the perturbative expansion in Eq. 2.13 as,

$$\begin{aligned}\Delta\left(\frac{\alpha_s}{2\pi}\right) &= 1 - \frac{\beta_0}{\epsilon}\frac{\alpha_s}{2\pi} + O\left(\frac{\alpha_s}{2\pi}\right) \\ &= \Delta^{(0)} + \Delta^{(1)}\frac{\alpha_s}{2\pi} + O\left(\frac{\alpha_s}{2\pi}\right).\end{aligned} \tag{2.18}$$

We are now able to generalise Eq. 2.16 to all antenna functions as follows,

$$\begin{aligned}X_{n,(k)}^l &= X_{n,(k)}^{l,\,\text{bare}} + \sum_{m=1}^{\min(l,k-1)} \Delta^{(m)} X_{n,(k-m)}^{l-m,\,\text{bare}} \\ &= X_{n,(k)}^{l,\,\text{bare}} + \Delta^{(1)} X_{n,(k-1)}^{l-1,\,\text{bare}} + \Delta^{(2)} X_{n,(k-2)}^{l-2,\,\text{bare}} + \ldots \\ &= \frac{1}{\left|\mathcal{M}_2^0\right|^2}\left[\left|\mathcal{M}_n^l\right|^2 + \Delta^{(1)}\left|\mathcal{M}_n^{l-1}\right|^2 + \Delta^{(2)}\left|\mathcal{M}_n^{l-2}\right|^2 + \ldots\right].\end{aligned} \tag{2.19}$$

Lastly, the $\mathcal{N}_X$ normalisation factor will also need to be generalised. It becomes,

$$\mathcal{N}_X = \frac{1}{2g_s^{2(n-2+l)}C_F}. \tag{2.20}$$

2.3.2 Colour Algebra and the Antenna Colour Decomposition

The processes represented by the X_n^l antenna functions are composed of colour-carrying particles, namely quarks and gluons. When the amplitudes $\mathcal{M}_n^l$ are evaluated, the kinematics and colour are not explicitly separate algebraically, but since we require the antenna functions to be purely kinematic, we need to decompose them into inner colour structures. To be able to decompose, we must first clarify the Algebra of colour [11, 13].

Quarks carry a fundamental colour index $i \in \{1, 2, 3\}$ while gluons carry an adjoint colour index $a \in \{1, ..., 8\}$. The interactions between them are governed by the $SU(3)$ generators T^a. Those generators are normalised by the trace relation $\mathrm{Tr}\left(T^a T^b\right) = T_R \delta^{ab}$, where by standard convention $T_R = 1/2$ [1, 2]. This is particularly useful since when the amplitude is squared, the two colour generators must be contracted with one another, that is T^a and $(T^a)^\dagger$, and since the generators are Hermitian, $(T^a)^\dagger = T^a$.

[6] These identities are explored further in Appendix [ref to the identities in an Appendix].

[7] In QCD terminology, "colour charge" (red, green and blue) refers to the fundamental quantum numbers carried by individual quarks and gluons, analogous to positive or negative electric charges. The "strong charge", more formally known as the strong coupling constant (g_s, where $\alpha_s = g_s^2/(4\pi)$), refers to the universal strength of the interaction between those colour-charged particles.

Hence, summing over all colour we get,

$$\sum_{i_1,i_2,a} (T^a)_{i_1 i_2} (T^a)_{i_2 i_1} = \sum_a \mathrm{Tr}\,(T^a T^a) = \sum_a T_R \delta^{aa}. \tag{2.21}$$

For a fixed a, the trace equals T_R. Summing this result over all $N^2 - 1$ adjoint indices we get,

$$\sum_a T_R \delta^{aa} = \frac{1}{2}\left(N^2 - 1\right). \tag{2.22}$$

We now also introduce the Casimir invariants: the fundamental and adjoint Casimir invariants, C_F and C_A. These represent, respectively, the total colour charge squared of the quark and of the gluon. They are defined as,

$$\begin{aligned} C_F &= \frac{N^2 - 1}{2N} \\ C_A &= N. \end{aligned} \tag{2.23}$$

The invariants are particularly useful to normalise the two hard quarks' colour from the antenna functions, by taking out a factor of $2C_F$. This factor equals,

$$2C_F = N - \frac{1}{N} \tag{2.24}$$

which will be a normalisation used throughout the present work.

In loop processes, colour can flow in different ways depending on the virtual particles inside the loops. So, instead of only simplifying using the fundamental Casimir invariant, we must also decompose the antenna structures into their distinct colour components. These are,

- The leading colour component (N): these come from planar Feynman diagrams (where virtual gluons are exchanged without crossing over each other). This component is represented by the base antenna in standard notation, e.g. A_3^1.
- The subleading colour component ($1/N$): these come from non-planar diagrams (where gluon lines do cross each other). This component is represented by the tilde antenna, e.g. $\tilde{A}_3^1$.
- The fermion loop (N_f): Instead of a virtual gluon, the loop can have a virtual quark-antiquark pair. This contribution depends strictly on the number of active flavours, N_f. This component is represented by the hat antenna, e.g. $\hat{A}_3^1$[11].

The direct expansion to Eq. 2.21 is to generalise to the four quark colour indices, instead of only two. That expansion is

$$\sum_a (T^a)_{i_1 i_2} (T^a)_{i_3 i_4} = \frac{1}{2}\left(\delta_{i_1 i_4}\delta_{i_2 i_3} - \frac{1}{N}\delta_{i_1 i_2}\delta_{i_3 i_4}\right). \tag{2.25}$$

It is the $SU(N)$ completeness relation[8] and is particularly useful for four-quark topologies like those described in Subsection 2.3.3 [1, 2].

A complete compilation of the QCD Feynman rules and the full colour trace derivations utilised by the package can be found in Appendix [ref to qcd appendix].

[8]This expression is also called the *colour Fierz identity*.

2.3.3 Antenna Families and T-Objects

Throughout this work the Born-level process under study is $\gamma^* \to q\bar{q}$. This process is represented at higher perturbative orders by the A, B and C antenna families. Antenna families are designations we use to identify the type of process that underlies a certain function. Consider the A_2^1 antenna: from our previous description, this antenna is a quark-antiquark antenna with a gluon loop. A_3^0 is a part of that same antenna family, the A family, those quark-antiquark antennae where gluons are exchanged or produced as final-state particles, but does not present with any loops. Instead, the gluon here is one of the now three final-state particles. Going to a higher level on the perturbative series, at NNLO, the quark-antiquark processes begin to allow for four final-state particle systems, in which the two unresolved partons may be other quark-antiquark pairs. This necessitates the creation of new antenna families since the aforementioned definition is now broken. We then designate the A_4^0 antenna for the $\gamma^* \to qgg\bar{q}$ process and the B_4^0 and C_4^0 antennae for the $\gamma^* \to q\bar{q}q\bar{q}$ [11]. We here require two different families because the process may present with either like (identical) or unlike-flavour pairs with the B antenna carrying the unlike-flavour case, often notated by $\gamma^* \to q\bar{q}q'\bar{q}'$, and the C antenna carrying the like-flavour case, denoted as $\gamma^* \to q\bar{q}q\bar{q}$. This last process, due to its topology, requires symmetrisation.

To compute these two antenna functions we start by writing the four-quark tree-level amplitude as a sum of two flavour-flow sub-amplitudes as,

$$\mathcal{M}_{q\bar{q}q\bar{q}} = \mathcal{M}_D + \mathcal{M}_E \tag{2.26}$$

where $\mathcal{M}_D$ is the direct and $\mathcal{M}_E$ is the exchanged [9] assignment amplitude.

The B_4^0 antenna, where the flavours are non-identical, so only the direct assignment amplitude is used. The antenna is built from self-interference,

$$B_4^0 = \mathcal{N}_X \frac{\mathcal{M}_D \mathcal{M}_D^*}{|\mathcal{M}_2^0|^2}. \tag{2.27}$$

On the other hand, C_4^0, where the flavours are identical, the two subamplitudes must interfere. The antenna is given as [1, 11],

$$C_4^0 = \mathcal{N}_X \frac{\mathcal{M}_D \mathcal{M}_E^* + \mathcal{M}_E \mathcal{M}_D^*}{|\mathcal{M}_2^0|^2}. \tag{2.28}$$

Though not deeply included in the present work, there are two other antenna types: there are quark-gluon antennae, which are organised under the D and E families, and there are gluon-gluon antennae, which are organised under the F, G and H families. As the names suggest, these types represent different Born-level processes. The first represents the boson $\to qg$ process, and the second the boson $\to gg$ process. Remarkably, these processes do not neatly fit into the Standard Model of Particle Physics (SM), as the vertices required to make a virtual photon split into either of those pairs does not exist inside the model. The implementation of those antenna functions, then, requires the use of Effective Field Theories (EFT). For the quark-gluon antennae the EFT used is SUSY and the process does not start in a virtual photon, but rather in a virtual neutralino. This neutralino splits into a gluon-gluino pair, as per SUSY and the gluino is then effectively considered an up-quark, hence describing a quark-gluon pair [19]. The gluon-gluon antennae require the use of the Higgs to gluon pair EFT framework. Here, the SM Feynman

[9] The words *direct* and *exchange* denote the way in which the like- and unlike-flavour quarks are paired amongst each other. When all flavours are identical, the pairing may be blind, since either of the quarks pairing with either of the antiquarks results in the same pair. If the flavours are not alike, however, the pairing order matters and must be considered, hence the use of the *exchanged* amplitude.

diagram describing a Higgs boson decaying into a top-quark triangle and then each of the other two vertices of that triangle emitting a gluon gets its intermediate triangle step compacted into the vertex, hence creating a $H \longrightarrow gg$ process and, hence, a gluon-gluon antenna [11, 20, 21]. These processes are describing in further detail in Appendix [ref to SUSY and HiggsEFT explanation appendix].

As described in Subsection 2.3.2, antenna functions are purely kinematic. By taking all components, we are able to construct a process-wide kinematics- and colour-conscious object. These are the $\mathcal{T}$-objects. These objects are defined, before integration and hence notated by T, conceptually as [10] ,

$$T^{2(k+1)}_{\text{topology}} = (\text{colour/flavour weight}) \times T^2_{q\bar{q}} \times (\text{colour-weighted sum of antennae at } k\text{-order}). \tag{2.29}$$

This structure creates a very elegant structure for the A-type antenna relative T-objects. It goes like,

$$T^{2(k+1)}_{\text{topology}} = 2C_F T^2_{q\bar{q}} \sum_c F_c A^l_{n,(k),c} \tag{2.30}$$

where c denotes the component (leading, subleading, fermion loop, etc), F_c is the colour weight, that is N, $1/N$, N_f, etc, and $A^l_{n,(k),c}$ is the relative component antenna with n final-state particles and l loops. For this antenna type, this definition needs two extensions at NNLO that do not cleanly fit into an algorithmic formula:

- The A^1_3 antenna set represents a particularly interesting process. Unlike in lower order loop processes, here two very distinct kinds of Feynman diagram appear: those in which the loop and the emission of the final-state gluon are completely separate events, and those in which the loop involves all three external legs. The first kind of diagram maps exactly a A^1_2 antenna topology followed by a A^0_3 topology: first the two hard quarks exchange a gluon and then one of the hard quarks emits a final-state gluon[11]. Since this behaviour is not singular to the A^1_3 antenna, it must be subtracted from the bare result to retain the antenna's singular behaviour. When we then construct back the T-object, the $A^1_2 A^0_3$ term has to be added back on such that the process is considered in its totality [11].

- The A^2_2 antenna set, also at NNLO, is the first time we find a double-loop topology. Here the extension required is not a subtraction of some kind of behaviour, but rather an addition of a new one. This set is, in reality, composed of two different processes: the first follows the standard rule closely, as it produces a leading, a subleading and a fermion loop component. The second, however, has a singular antenna which describes the self-interference of the one-loop correction with itself, now that the system presents with two loops. This antenna is notated by $\breve{A}^2_2$. It is also the only genuinely square (not cross) term at NNLO. This means that it requires a normalisation of $(2C_F)^2$ [11].

As mentioned before Eq. 2.30, the equation is only valid for T-objects constructed from A-type antenna functions. In this work we consider all quark-antiquark antennae up to NNLO. This includes B^0_4 and C^0_4. These antennae populate the $T^6_{q\bar{q}q'\bar{q}'}$ and $T^6_{q\bar{q}q\bar{q}}$ objects, respectively.

$T^6_{q\bar{q}q'\bar{q}'}$ is built in the same way as other single-topology antenna T-objects, like $T^4_{q\bar{q}g}$, with the only difference being that here all quark flavours are summed over, requiring the inclusion of a $(N_f - 1)$ term,

[10] The T-objects are commonly notated by T^2, T^4 and T^6, depending on their perturbative order: LO, NLO and NNLO, respectively. That is the same as using twice our previous k-index, introduced in 2.3.1, plus one. This is the notation used in [22] and the one that will be used for the rest of the present work.

[11] This is merely a conceptual, diagrammatic interpretation, as the actual process is much less sequential than this description suggests.

constructing,

$$T^6_{\bar{q}\bar{q}q'\bar{q}'} = 2C_F T^2_{q\bar{q}} (N_f - 1) B^0_4. \tag{2.31}$$

The identical-flavour quark-antiquark-quark-antiquark final state is described by the sum of antennae for non-identical-flavour and for identical-flavour-only [11]. Using Eq. 2.25 in the interference term, with the identity piece vanishing by tracelessness [12], we can describe the identical-flavour T-object as,

$$T^6_{q\bar{q}q\bar{q}} = 2C_F T^2_{q\bar{q}} \left[B^0_4 - \frac{1}{N} C^0_4 \right] \tag{2.32}$$

importantly without the inclusion of the quark-flavour sum term $(N_f - 1)$.

2.3.4 Phase-Space Factorisation

The previous Subsections focused on the antenna functions as objects: how they are built, combined, normalised, etc. But the antenna functions as described in Subsection 2.3.1 are the singular factor $\mathcal{S}_j$, as described in Eq. 2.15, up to some normalisation factor. The antenna functions, then, as part of $d\sigma^S$ must be integrated as, schematically, as described in Eq. 2.3,

$$\int_1 d\sigma^S \tag{2.33}$$

where $\int_1$ denotes the integral over the unresolved parton phase-space. This integral does not only denote the integral over the single unresolved parton, alone, on two counts: first the fact that the integral must cover all unresolved partons which, for three-final-particle antennae is only one, but for four-final-particle antennae it becomes two. Second, the hard partons and their interactions with the unresolved partons must also be integrated over. The integral becomes over the n-final-state-particle antenna phase space. This makes,

$$\int_1 d\sigma^S \sim \int d\Phi_{ijkl\ldots} X^l_n \tag{2.34}$$

up to normalisation [11].

The n-final-state-particle antenna phase space $\Phi_{ijkl\ldots}$ [13] is not the standard n-particle phase space as in Φ_3 or Φ_4, it is, rather, the antenna's phase space, associated with the unresolved-partons emission between the two hard radiators, it is the unresolved-parton phase space of the antenna. It has no independent parametrisation and is only accessed by factorising the known n-particle phase space [11, 23]. Since, through the recursive factorisation of the Lorentz-invariant phase space property [14] [8], the phase space factorises, we are able to denote that the n-particle phase-space equals the $(n - n_u)$-particle, where n_u is the number of unresolved partons, phase-space times the antenna phase-space for n final-state particles [15]. For our case, two useful identities are produced for this formula [11],

$$d\Phi_3 = d\Phi_2 d\Phi_{ijk}$$

$$d\Phi_4 = d\Phi_2 d\Phi_{ijkl}. \tag{2.35}$$

[12] This process is derived completely in Appendix [ref to the derivation].

[13] This phase space is notated by Φ_{ijk} for three-final-state-particle antennae like A^1_3 and Φ_{ijkl} for four-parton-final-particle antennae like A^0_4.

[14] This property is discussed further in Appendix [ref to relevant appendix].

[15] This mechanism, as described, works at $n_u \in \{1, 2\}$. The perturbative order above, N^{3}LO, presents more complex identities for which this blueprint may not work. It also does not fit when $n_u = 0$. That case is discussed in further detail in Subsubsection 2.3.4.

Since $d\Phi_n$ is perfectly factorised [8], and the X_n^l antenna functions are explicitly built to be a function of the antenna's internal invariants alone and to have no dependence on how the reduced system is oriented [11], the two phase spaces are integrated separately, with the two-particle phase-space integral producing Φ_2 directly. The integrated antenna $\mathcal{X}_n^l$ is then computed as,

$$\mathcal{X}_n^l \sim \int d\Phi_{ijkl\ldots}\, X_n^l = \frac{1}{\Phi_2} \int d\Phi_n\, X_n^l \tag{2.36}$$

which is how the phase-space prefactor appears in $\mathcal{N}(\epsilon, k)$, as described in Section 2.2.

Hence, finally, the integrated antenna $\mathcal{X}_n^l$ is given generally[16] as, taking the aforementioned overall $\mathcal{N}$ normalisation factor [11, 23],

$$\mathcal{X}_n^l = \mathcal{N}(\epsilon, k) \int d\Phi_n\, X_n^l. \tag{2.37}$$

A Note On $n_u = 0$ Cases

There are two antenna types that are exempt from this rule: A_2^1 and A_2^2, as they do not present with unresolved partons, but rather only with loops, and are completely virtual antennae. These antennae are only integrated over the loop momenta rather over the phase space, thus not requiring normalisation using the two-particle phase-space [24].

Implications on the T-objects

With this definition of the integrated antenna $\mathcal{X}_n^l$, we are now also able to introduce the often used integrated T-object as,

$$\mathcal{T}_{\text{topology}}^{2(k+1)} = 2C_F T_{q\bar{q}} \sum_c F_c \mathcal{X}_{n,(k),c}^l \tag{2.38}$$

written schematically from Eq. 2.30, though the structure may change with the antenna type used to define the relevant T-object. This equation describes that the integrated T-object, $\mathcal{T}_{\text{topology}}^{2(k+1)}$ is defined using the integrated antennae $\mathcal{X}_n^l$, integrated over their respective phase-spaces rather than blindly integrating the unintegrated T-object directly [11, 22].

2.3.5 NLO and NNLO Subtraction Structure

So far, throughout the present Section 2.3, we have discussed what antenna functions are and represent, how they are normalised, what conventions are used when using them and the overall objects we use the antenna functions to compute, the T-objects. What we have not yet done, and hence not closed the section's loop is to explain why we require both the antenna functions and the T-objects. We require them, as hinted at in Subsection 2.1.1, to compute the $d\sigma^S$ differential subtraction counterterm used in the antenna subtraction scheme, as schematically described in Eq. 2.3 [8].

The main reason behind the antenna subtraction scheme is to create a finite differential prediction tool (the differential cross-section) for some F_n infrared-safe quantity, or observable, for n final-state particles, which here acts as a distribution. The terms of the subtraction formula must, then, carry their own infrared-safe quantities through the process. As a matter of example, the differential (F_n-dependent)

[16] This definition is general for antenna functions that are already cleanly and completely given in terms of Mandelstam invariants and ϵ. This is only automatically true for antenna where $l = 0$. When $l \geq 0$, which happens for all loop antennae by definition, a prior loop-momentum integration step is necessary. This step is explored further in Section 2.4.

NLO finite cross-section is given by,

$$d\sigma^{\text{NLO}} = \int_{m+1}\left(d\sigma^R F_{m+1} - d\sigma^S F_m\right) + \int_m \left(d\sigma^V + \int_1 d\sigma^S\right) F_m. \tag{2.39}$$

where $m = 2$ and is the number of final-state particles at Born-level [1].

Since the observable considered throughout this work, the hadronic R-ratio, is fully-inclusive, making no distinction between resolved and unresolved final states at any multiplicity, the measurement distribution function F_n equals one identically, for every n. The cross-section, in our case, is therefore no longer a differential prediction in F_n but rather reduces to the plain, fully-summed total. This enables us to write,

$$\sigma^{\text{NLO}} = \int_{m+1}\left(d\sigma^R - d\sigma^S\right) + \int_m \left(d\sigma^V + \int_1 d\sigma^S\right) \tag{2.40}$$

which maps directly to Eq. 2.3 and represents the scheme at NLO.

The scheme looks different at NNLO, simply due to the larger number of possible corrections. Instead of only real and virtual cases, this order's correction receives contributions where the two additional orders of α_s are realised as two extra real partons (RR), one extra real parton and one loop (RV), or two loops with no extra partons (VV). The expression then expands to meet these new terms to,

$$\sigma^{\text{NNLO}} = \int_{m+2}\left(d\sigma^{RR} - d\sigma^S\right) + \int_{m+1}\left(d\sigma^{RV} - d\sigma^T\right) + \int_m \left(d\sigma^{VV} + \int_1 d\sigma^T + \int_2 d\sigma^S\right) \tag{2.41}$$

where instead of only one, there are two distinct subtraction counterterms. These are $d\sigma^S$, which subtracts the singularities of $d\sigma^{RR}$, at both single- and double-unresolved level, and $d\sigma^T$, which subtracts the remaining single-unresolved singularities of $d\sigma^{RV}$ [11]. This inadvertently adds another level of complexity for NNLO: the KLN theorem guarantees only that the fully inclusive sum over degenerate configurations is finite, but says nothing about any particular piece of a decomposition being finite on its own. With the split including multiple subtractions at the same time, we are implying that each of these differences is separately, pointwise finite, everywhere in its own phase space.

As hinted at throughout this text but not yet stated explicitly: the subtraction terms are exact, not merely asymptotic, matches to the original cross-section terms. This follows from two facts specific to this process. First, because the antenna functions are built as exact, unique expressions (Eq. 2.16), and the sum in Eq. 2.15 runs over all unresolved pairs, of which there is only one, since the Born process $\gamma^* \to q\bar{q}$has a single colour-connected pair. Second, because the Born factor $|\mathcal{M}_2^0|^2$ depends only on the conserved quantity q^2, and therefore takes the same value whether evaluated on the exact real-emission momenta or on the reconstructed momenta used in the subtraction term. Together, these mean $d\sigma^S$ and $d\sigma^T$ coincide identically with $d\sigma^R$, $d\sigma^{RR}$, and $c\sigma^{RV}$ at every point in the phase space, for either perturbative order, not merely in the singular unresolved limit, as the general subtraction formalism only guarantees [8]. an example, σ^{NNLO}, from Eq.2.41, simplifies to,

$$\sigma^{\text{NNLO}} = \int_m \left(d\sigma^{VV} + \int_1 d\sigma^T + \int_2 d\sigma^S\right) \tag{2.42}$$

in which each of the terms is described by a different T-object, as described in Subsection 2.3.3. The objects then describe each of the cross-section terms directly, and a clear hierarchy of objects to cross-sections can be made looking at the number of unresolved partons in each of them. The expression then becomes,

$$\sigma^{\text{NNLO}} = \frac{1}{T^2_{q\bar{q}}} \left[\mathcal{T}^6_{q\bar{q}} + \mathcal{T}^6_{q\bar{q}g} + \left(\mathcal{T}^6_{q\bar{q}q\bar{q}} + \mathcal{T}^6_{q\bar{q}q'\bar{q}'} + \mathcal{T}^6_{q\bar{q}gg} \right) \right] \tag{2.43}$$

where the $1/T^2_{q\bar{q}}$ term is added to normalise the LO order to one, as $\sigma^{\text{LO}} = T^2_{q\bar{q}}$ and where $\mathcal{T}$ denotes the integrated form of a certain T-object, as described in Eq. 2.38.

The NLO case goes in the same direction exactly. Its cross-section becomes,

$$\sigma^{\text{NLO}} = \frac{1}{T^2_{q\bar{q}}} \left(\mathcal{T}^4_{q\bar{q}} + \mathcal{T}^4_{q\bar{q}g} \right). \tag{2.44}$$

Since the T-objects are described using the antenna functions, by necessity, both σ^{NLO} and σ^{NNLO} may also be given in terms of the respective antenna functions, after the normalisation procedures described throughout this text, as follows [11],

$$\begin{aligned}\sigma^{\text{NNLO}} = \left(N - \frac{1}{N}\right) \Bigg[& N \left(\mathcal{A}^1_2 \mathcal{A}^0_3 + \mathcal{A}^1_3 + \mathcal{A}^0_4 + \mathcal{A}^2_2 \right) - \frac{1}{N} \left(\mathcal{A}^1_2 \mathcal{A}^0_3 + \tilde{\mathcal{A}}^1_3 + \tilde{\mathcal{A}}^0_4 + \mathcal{C}^0_4 - \tilde{\mathcal{A}}^2_2 \right) \\ & + N_f \left(\hat{\mathcal{A}}^1_3 + \hat{\mathcal{A}}^2_2 \right) + \left(N - \frac{1}{N} \right) \breve{\mathcal{A}}^2_2 \Bigg] \\ \sigma^{\text{NLO}} = \left(N - \frac{1}{N}\right) & \left(\mathcal{A}^1_2 + \mathcal{A}^0_3 \right). \end{aligned} \tag{2.45}$$

2.4 Phase-Space Integration and IBP Reduction

In the previous sections we went over what an antenna function is, why they are necessary and how we use them. In Eq. 2.37 we defined the method we use to integrate the antenna functions X^l_n, to integrated antenna functions $\mathcal{X}^l_n$. These antennae are integrated over the phase-space and also, when loops are present, their loop momenta ℓ_r, as pointed out in Footnote 16.

Though integration has been made to seem the natural endpoint of this present work, and to an extent it is, there is still use among phenomenologists for the unintegrated antenna function X^l_n, namely when using Monte Carlo (MC) integrators. These integrators evaluate the phase space numerically point-by-point and require our unintegrated antennae to perform subtractions over local divergences, as described in Subsection 2.3.5 [11]. Since the MC generator throws random phase-space points, and the subtraction happens dynamically with each throw, the unintegrated antenna functions used must be given in terms of the exact same local kinematics as the real-emission matrix elements, the Mandelstam invariants s_{ij}. For this reason, we need to perform the loop-momentum integral before using these antennae in these integrators, if $l \geq 1$[17].

From Eq. 2.36, we have seen that through phase-space factorisation, the antennae are integrated over the antenna phase space Φ_X. If required, they are also integrated over their loop momenta. The general expression used to describe this integration setup is,

$$\mathcal{X}^l_n \sim \int \left(\prod_{r=1}^{l} d^d \ell_r \right) d\Phi_X \, \frac{N_s(\ell_r, p_a)}{\prod_{j=1}^{N} D_j^{a_j}(\ell_r, p_a)}, \quad d = 4 - 2\epsilon \tag{2.46}$$

where ℓ_r denote the loop momenta, and the set p_a consists of a fixed mapped hard-radiator momenta

[17] Note that in standard antenna language, the term *unintegrated* already refers to phase-space integration, over Φ_X.

$\tilde{p}_{\text{hard}}$ and the unresolved antenna momenta $p_{\text{unresolved}}$, the latter of which are integrated through Φ_X. N_s is a numerator built from scalar products, D_j are propagator or invariant denominators and a_j are integer powers. An algebraic expansion of the integrand of a single amplitude can generate multiple integrals with the same kinematic structure but different numerator and denominator powers. These integrals may then be organised into integral families, which fix the denominators and integration variables, and may then be systematically reduced. These families are denoted as the complete collection [23],

$$\mathcal{F}[\{D_j\};\{\ell_r\},d\Phi_X] = \{I(a_1,...,a_N)\}, \quad a_j \in \mathbb{Z} \tag{2.47}$$

with,

$$I(\vec{a}) = \int \left[\prod_{r=1}^{l} d^d\ell_r\right] d\Phi_X \, \frac{1}{D_1^{a_1}...D_N^{a_N}}, \quad \vec{a} = (a_1,...,a_N). \tag{2.48}$$

To reduce these families' terms, denoted as I here, is to write them in terms of a finite basis as,

$$I(\vec{a}) = \sum_{k=1}^{N_{\text{MI}}} c_k(\vec{a},\epsilon,\{s_{ij}\})M_k \tag{2.49}$$

where M_k are the master integrals (MI). In essence, it takes the problem of integrating those many family terms I and separates into two parts. In I, positive a_j indices denote propagator powers while non-positive indices account for absent denominators or irreducible scalar products in the numerator. The reduction separates the calculation into determining the family-specific coefficients c_k and evaluating the master integrals M_k, so that every required family member is expressed through a substantially smaller basis.

As stated previously, MC integrators require the unintegrated antenna functions in terms of the Mandelstam invariants s_{ij}, since they must evaluate the subtractions at a sampled physical external phase-space point p. The loop-level antenna required by an MC is ultimately evaluated as a function of the external invariants. Its calculation, however, involves tensor loop integrals with loop-momentum dependence, which must be reduced before the antenna can be evaluated at a sampled phase-space point. This reduction is performed through the Passarino-Veltmann (PaVe) reduction mechanism, described in detail in Subsection 2.4.1, which reduces the tensor loop integrals to linear combinations of scalar one-loop functions, usually denoted by A_0, B_0, ..., while keeping the phase-space variables and integration untouched, as [25],

$$\int d^d\ell \, \frac{\ell^{\mu_1}...\ell^{\mu_R}}{\prod_j D_j^{a_j}} \xrightarrow{\text{PaVe}} \sum_k c_k\big(\{s_{ij}\}\big)\,\{A_0,B_0,C_0,D_0,...\}_k. \tag{2.50}$$

To finally reach the integrated antenna functions $\mathcal{X}_n^l$ we do require integrating over the phase-space. For this integral we also use a reduction method: Integration By Parts (IBP) reduction, described in detail in Subsection 2.4.2. This method generates relations among scalar integrals in a common family through vanishing total derivatives, as [26],

$$0 = \int d^d\ell \, \frac{\partial}{\partial\ell^\mu}\left(v^\mu \frac{1}{\prod_j D_j^{a_j}}\right). \tag{2.51}$$

As mentioned in previous sections, emitted final-state particles are real and on shell. That condition is enforced in the phase-space through the use of Dirac-delta functions like $\delta^+\big(p_i^2\big)$. Note that these are not ordinary Feynman propagators. To bring the phase-space integration into the same algebraic framework as the loop integrals, we introduce the method of reverse unitarity to represent these Dirac-delta functions as cut propagators, through the discontinuity relation as [23],

$$2\pi i\, \delta^+\!\left(p_i^2\right) = \left[\frac{1}{p_i^2 - i0} - \frac{1}{p_i^2 + i0}\right]_+ . \tag{2.52}$$

where both + signs, in δ^+ and $]_+$, represent the positive-energy prescription.

The family now contains ordinary propagators from virtual loops and cut propagators representing the unresolved phase-space. The IBP identities can then reduce the resulting combined cut-integral family to a finite set of MI.

After UV renormalisation and evaluation of the MI, the result is expanded as a Laurent series in ϵ, making the remaining explicit IR poles visible order by order [11, 9].

2.4.1 PaVe Reduction

At one-loop, amplitudes produce tensor integrals similar to those in Eq. 2.50 [25]. With general massive[18] denominator defined as,

$$D_0 = \ell^2 - m_0^2 + i0, \quad D_i = (\ell + r_i)^2 - m_i^2 + i0 \tag{2.53}$$

the tensor integral takes the form that follows,

$$I_N^{\mu_1 \dots \mu_R} = \mu^{4-d} \int \frac{d^d\ell}{i\pi^{d/2}} \frac{\ell^{\mu_1} \dots \ell^{\mu_R}}{D_0 D_1 \dots D_{N-1}} \tag{2.54}$$

where R is the tensor rank, r_i are the combinations of external momenta and m_i is the mass associated with the i-th propagator. The reduction is carried out in $d = 4 - 2\epsilon$ dimensions, so that,

$$g_{\mu\nu} g^{\mu\nu} = d. \tag{2.55}$$

PaVe reduces the Lorentz indices in the numerator $\ell^\mu \ell^\nu \ell^\rho \dots$ terms and expresses them in terms of external momenta p_i^μ, the metric tensor $g_{\mu\nu}$ and scalar one-loop integrals, through Lorentz covariance [25].

For a rank-one N-point integral, the only available vectors are the independent external momenta, such that,

$$I_N^\mu = \sum_{i=1}^{N-1} p_i^\mu C_i \tag{2.56}$$

where C_i are scalar coefficient functions. For rank-two, the relation becomes,

$$I_N^{\mu\nu} = \sum_{i,j=1}^{N-1} p_i^\mu p_j^\nu C_{ij} + g^{\mu\nu} C_{00} \tag{2.57}$$

[18] Throughout this subsection, the general and massive case will be considered for completeness. The rest of the present work, however, heavily leans towards the massless regime. In that regime the internal masses vanish and the physical external partons satisfy $p_a^2 = 0$. The shifted momenta r_i, introduced in Eq. 2.53, entering propagators are combinations of external momenta and need not be lightlike.

and for rank-three,

$$I_N^{\mu\nu\rho} = \sum_{i,j,k=1}^{N-1} p_i^\mu p_j^\nu p_k^\rho C_{ijk} + \sum_{i=1}^{N-1} \left(g^{\mu\nu} p_i^\rho + g^{\mu\rho} p_i^\nu + g^{\nu\rho} p_i^\mu\right) C_{00i} \tag{2.58}$$

and so on for higher ranks.

The reduction mechanism works because scalar products involving the loop momentum can be rewritten using the propagator denominators as,

$$2\ell \cdot r_i = D_i - D_0 - r_i^2 + m_i^2 - m_0^2 \tag{2.59}$$

which lets us cancel denominator terms, through a method like the following example set at $i = 1$,

$$\frac{\ell \cdot r_1}{D_0 D_1} = \frac{1}{2}\left(\frac{1}{D_0} - \frac{1}{D_1} + \frac{1}{D_0 D_1}\left(-m_0^2 + m_1^2 - r_1^2\right)\right) \tag{2.60}$$

thereby rewriting loop-momentum scalar products in terms of denominators and scalar integrals.

While using this reduction scheme, we might also encounter Gram matrices, defined as,

$$G_{ji} = 2p_j \cdot p_i. \tag{2.61}$$

Contracting a general rank-one integral, as described in Eq. 2.56, with p_j gives,

$$p_j \cdot I_N = \sum_i (p_j \cdot p_i) C_i = \frac{1}{2}\sum_i G_{ji} C_i. \tag{2.62}$$

If we now define $R_j \equiv p_j \cdot I_N$, we are then able to compute the C_i coefficients in the Lorentz decomposition by taking[19],

$$G_{ji} C_i = 2R_j \Rightarrow C_i = 2\left(G^{-1}\right)_{ij} R_j. \tag{2.63}$$

Rank-One Bubble Example

To illustrate the method, let us consider the rank-one bubble,

$$B^\mu\left(p; m_0^2, m_1^2\right) = \mu^{4-d} \int \frac{d^d\ell}{i\pi^{d/2}} \frac{\ell^\mu}{D_0 D_1} \tag{2.64}$$

where the denominators follow Eq. 2.53. By Lorentz covariance we are able to write that,

$$B^\mu = p^\mu B_1 \tag{2.65}$$

which once contracted with p_μ makes,

$$p^2 B_1 = p_\mu B^\mu = \mu^{4-d} \int \frac{d^d\ell}{i\pi^{d/2}} \frac{\ell \cdot p}{D_0 D_1}. \tag{2.66}$$

[19]The inversion in Eq.2.63 assumes a non-singular Gram matrix. Exceptional kinematics may require a limiting procedure or an alternative reduction method.

Using the relation in Eq. 2.59, where we consider $r_1 = p$, we get,

$$p^2 B_1 = \frac{1}{2}\left[A_0\left(m_0^2\right) - A_0\left(m_1^2\right) + \left(m_1^2 - m_0^2 - p^2\right) B_0\left(p^2; m_0^2, m_1^2\right)\right] \tag{2.67}$$

and therefore[20],

$$B^\mu = p^\mu \frac{A_0(m_0^2) - A_0(m_1^2) + (m_1^2 - m_0^2 - p^2)\, B_0}{2p^2}. \tag{2.68}$$

Here, A_0 and B_0 are the scalar one-loop integrals mentioned throughout. These two types can be defined as,

$$\begin{aligned} A_0\left(m^2\right) &= \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{1}{\ell^2 - m^2 + i0} \\ B_0\left(p^2; m_0^2, m_1^2\right) &= \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{1}{D_0 D_1}. \end{aligned} \tag{2.69}$$

2.4.2 IBP Identities and the Laporta Algorithm

IBP Identities

The integral family has already been defined in Eq. 2.48. This definition allows for the factorisation of the integral into a loop and a phase-space integral term. That factorisation results in,

$$I(\vec{a}) = \int d\Phi_X\, \mathcal{I}(\vec{a}; \{D_1, ..., D_N\}), \quad \mathcal{I}(\vec{a}; \{D_1, ..., D_N\}) = \int \prod_{r=1}^{l} d^d \ell_r \frac{1}{D_1^{a_1} ... D_N^{a_N}}. \tag{2.70}$$

The overall normalisation factors are irrelevant to the identities and will, therefore, be suppressed throughout the following explanation. The denominators may generally be defined as,

$$D_j = \left(\sum_{r=1}^{l} c_{jr} \ell_r + r_j\right)^2 - m_j^2 + i0 \tag{2.71}$$

where r_j is built from momenta held fixed during the loop integration, including the mapped hard partons and unresolved antenna momenta. Take one loop momentum ℓ_s, for $s = 1, ..., l$, and one vector v^μ from the available loop and external momenta. Dimensional regularisation allows us to set a total derivative to zero inside $\mathcal{I}$, as [23, 26],

$$0 = \int \prod_{r=1}^{l} d^d \ell_r \frac{\partial}{\partial \ell_s^\mu}\left[v^\mu \frac{1}{D_1^{a_1} ... D_N^{a_N}}\right] \tag{2.72}$$

as previously introduced in Eq. 2.51. In dimensional regularisation, analytic continuation justifies the vanishing of the surface term at infinity, leaving an exact relation among regulated integrals[21].

Expanding the derivative gives,

$$0 = \int \prod_{r=1}^{l} d^d \ell_r \left[\frac{\partial_{\ell_s} \cdot v}{\prod_j D_j^{a_j}} - \sum_{j=1}^{N} a_j \frac{v^\mu \partial_{\ell_s^\mu} D_j}{D_1^{a_1} ... D_j^{a_j+1} ... D_N^{a_N}}\right]. \tag{2.73}$$

[20] The result shown in Eq. 2.68 assumes generic kinematics with $p^2 \neq 0$. Exceptional cases are obtained through an appropriate limit or alternative reduction.

[21] This method is explored further and completely in Appendix [ref to this].

For these quadratic denominators D_j, differentiation with respect to ℓ_s^μ is linear in momenta,

$$\partial_{\ell_s^\mu} D_j = 2c_{js}\left(\sum_r c_{jr}\ell_r + r_j\right)_\mu \tag{2.74}$$

and therefore, $v^\mu \partial_{\ell_s^\mu} D_j$ is a scalar product. As scalar products may be expressed in terms of chosen denominators and irreducible scalar products, the latter are represented by negative family indices a_j, while zero indices denote absent denominators, we are able to write schematically,

$$\sum_{\vec{b}} \rho_{\vec{b}}\left(\vec{a}, \epsilon, \{s_{ij}\}\right) \mathcal{I}(\vec{b}) = 0 \tag{2.75}$$

where each vector $\vec{b}$ labels a family member obtained from $\vec{a}$ through index shifts in one or more denominator powers. $\rho_{\vec{b}}$ is the corresponding coefficient in the IBP relation.

The method may be made easier to visualise with an example. Consider a one-loop bubble, akin to that studied in the previous Subsection 2.4.1. Take,

$$\mathcal{I}(a_0, a_1) = \int d^d\ell \, \frac{1}{D_0^{a_0} D_1^{a_1}} \tag{2.76}$$

where,

$$D_0 = \ell^2 - m_0^2, \quad D_1 = (\ell + p)^2 - m_1^2. \tag{2.77}$$

We now choose $v^\mu = \ell^\mu$, and build and expand the relation from Eq. 2.51 as,

$$\begin{aligned} 0 &= \int d^d\ell \, \frac{\partial}{\partial \ell^\mu}\left(\frac{\ell^\mu}{D_0^{a_0} D_1^{a_1}}\right) \\ &= d\mathcal{I}(a_0, a_1) - 2a_0 \int d^d\ell \, \frac{\ell^2}{D_0^{a_0+1} D_1^{a_1}} - 2a_1 \int d^d\ell \, \frac{\ell \cdot (\ell + p)}{D_0^{a_0} D_1^{a_1+1}}. \end{aligned} \tag{2.78}$$

We can now use,

$$\ell^2 = D_0 + m_0^2 \tag{2.79}$$

$$2\ell \cdot (\ell + p) = D_0 + D_1 + m_0^2 + m_1^2 - p^2 \tag{2.80}$$

and finally get the explicit shifted-index identity,

$$\begin{aligned} 0 = &(d - 2a_0 - a_1)\mathcal{I}(a_0, a_1) \\ &- 2a_0 m_0^2 \mathcal{I}(a_0 + 1, a_1) \\ &- a_1 \mathcal{I}(a_0 - 1, a_1 + 1) \\ &- a_1\left(m_0^2 + m_1^2 - p^2\right)\mathcal{I}(a_0, a_1 + 1). \end{aligned} \tag{2.81}$$

Reverse Unitarity and Cut Integrals

In the previous subsubsection, we have derived the IBP identities relative to $\mathcal{I}$. As described in Eq. 2.70, the entire integrated antenna is not merely $\mathcal{I}$, but rather,

$$I(\vec{a}) = \int d\Phi_X \, \mathcal{I}(\vec{a}). \tag{2.82}$$

The unresolved phase-space measure is not originally an ordinary loop integral, however. Instead, it has on-shell constraints, which have to be considered and transformed before $I(\vec{a})$ is reducible via the present method.

As described in Subsection 2.3.4, phase-space factorisation isolates the antenna measure through $d\Phi_{m+r} = d\Phi_m d\Phi_X$, or in the two-hard-radiator final-final case we are considering throughout the present work, $d\Phi_X = d\Phi_n / d\Phi_2$, for n final-state particles. That n-particle phase space is given generally as,

$$d\Phi_n(q : p_1, ..., p_n) = (2\pi)^d \delta^{(d)}\left(q - \sum_{i=1}^{n} p_i\right) \prod_{i=1}^{n} \frac{d^d p_i}{(2\pi)^{d-1}} \, \delta^+\left(p_i^2 - m_i^2\right) \tag{2.83}$$

where, as described in the start of the current section, the δ-function imposes the mass-shell condition for the i-th final-state particle, while the superscript + selects its positive-energy solution. This may be written as,

$$\delta^+\left(p_i^2 - m_i^2\right) = \theta\left(p_i^0\right) \, \delta\left(p_i^2 - m_i^2\right) \tag{2.84}$$

This δ^+-function may be reexpressed as a discontinuity as follows [23, 27],

$$2\pi i \, \delta^+\left(p_i^2 - m_i^2\right) = \left[\frac{1}{p_i^2 - m_i^2 - i0} - \frac{1}{p_i^2 - m_i^2 + i0}\right]_+ \tag{2.85}$$

where the $\pm i0$ terms specify the propagator prescriptions. The subscript + represents the positive-energy cut selection.

These objects are called *cut propagators*, because they come about from cutting a propagator line, thus putting the represented particle on-shell, as a real final-state particle. Algebraically, it also allows phase-space constraints to be written in the same denominator language as virtual propagators. A cut propagator associated with a required final-state particle must be retained during the reduction and sectors in which such a cut is absent do not represent the physical phase-space integral and thus, vanish [23]. Using the discontinuity relation to refactor the δ-functions, we are able to rewrite the integrated antenna expression schematically as,

$$\mathcal{X}_n^l \sim \int \left[\prod_{r=1}^{l} d^d \ell_r\right] \left[\prod_{u} d^d p_u\right] \frac{N_s(\ell_r, p_u; \tilde{p}_{\text{hard}})}{\prod_\alpha D_{v,\alpha}^{a_\alpha} \prod_\beta D_{c,\beta}^{a_\beta}} \tag{2.86}$$

where $D_{v,\alpha}$ are ordinary virtual propagators, $D_{c,\beta}$ are cut propagators, $\tilde{p}_{\text{hard}}$ remains fixed as the hard-radiator combined momenta and p_u are integrated unresolved momenta.

After rewriting the integral in this way, the total derivative introduced in the previous Subsection 2.4.2 may now be taken with respect to any loop momentum ℓ_r^μ or integrated unresolved momentum p_u^μ. This process produces relations involving both virtual and cut denominators. This then allows us to reduce loop and phase-space integrals simultaneously using IBP reduction [23, 27].

As an example, let us now consider the massless three-particle phase-space Φ_3. It may be described as,

$$d\Phi_3(q; p_1, p_2, p_3) \propto d^d p_1 d^d p_2 d^d p_3\, \delta^+\left(p_1^2\right)\, \delta^+\left(p_2^2\right)\, \delta^+\left(p_3^2\right)\, \delta^{(d)}(q - p_1 - p_2 - p_3). \tag{2.87}$$

We start by using momentum conservation to eliminate p_3,

$$p_3 = q - p_1 - p_2. \tag{2.88}$$

This first step eliminates the rightmost δ-function. Then, using reverse unitarity, each of the remaining δ-functions may be rewritten as cut propagators. Schematically, it gives,

$$d\Phi_3 \sim d^d p_1 d^d p_2 \frac{1}{[p_1^2]_c [p_2^2]_c [(q - p_1 - p_2)^2]_c} \tag{2.89}$$

with

$$\left[\frac{1}{p^2}\right]_c \equiv \frac{1}{2\pi i}\left[\frac{1}{p^2 - i0} - \frac{1}{p^2 + i0}\right]_+ . \tag{2.90}$$

Laporta Algorithm

At NLO and NNLO these reduction systems become very complex and their solution ever more time-intensive. The Laporta algorithm is a systematic-elimination method that uses IBP relations to express more complicated integrals in terms of a smaller set of unreduced ones (master integrals) [28].

The algorithm first classifies the family members into sectors. These sectors are defined as,

$$\theta_j \equiv \theta(a_j) = \begin{cases} 1, & a_j > 0, \\ 0, & a_j \leq 0, \end{cases} \qquad \vec{\theta} = (\theta_1, ..., \theta_N). \tag{2.91}$$

After fixing an integral family and identifying its admissible sectors, the Laporta algorithm assigns an ordering to the resulting integrals according to their complexity. This hierarchy is based on the number of propagators present t, the excess denominator power r, and the numerator complexity encoded through negative indices s[22]. These quantities are defined using each integral's a-index, taken from the family's $\vec{a}$ vector, as previously defined. They are, generally,

$$t = \sum_{j=1}^{N} \theta(a_j), \tag{2.92}$$

$$r = \sum_{j=1}^{N} \max(a_j - 1, 0), \tag{2.93}$$

$$s = \sum_{j=1}^{N} \max(-a_j, 0). \tag{2.94}$$

Here, $\theta(a_j)$ records whether denominator D_j is present. The hierarchy is defined by comparing t first, then r and finally s. The algorithm uses this chosen hierarchy to select, within each IBP relation, the most complex integral to eliminate first.

As an example, consider an arbitrary family defined as $\mathcal{F}[\{D_1, D_2, D_3\}; d\Phi]$, with the denominators

[22]The present t, r and s metrics are not to be taken as unique Laporta orderings, as these vary by implementation. Instead, they are illustrative objects.

chosen as $D_1 = s_{12}$, $D_2 = s_{13}$ and $D_3 = s_{23}$. We may take the integrals below, all members of this family, and organise them using their respective sector vectors as follows,

$$I_1 = \int d\Phi \, \frac{s_{13}}{s_{12}} \Rightarrow I_1 = I(1, -1, 0), \tag{2.95}$$

$$I_2 = \int d\Phi \, \frac{2}{s_{12}s_{13}s_{23}} \Rightarrow I_2 = 2I(1, 1, 1), \tag{2.96}$$

$$I_3 = \int d\Phi \, \frac{1}{s_{12}^3} \Rightarrow I_3 = I(3, 0, 0) \tag{2.97}$$

where $d\Phi$ is the common phase-space measure. Importantly, two integrals may share a sector if they contain the same denominators, independently of those denominators' powers. Also, only positive indices denote denominator presence, meaning that integrals I_1 and I_3 in the example above are members of the same sector, $I_1, I_3 \in (1, 0, 0)$. In our example, only two sectors appear: $(1, 0, 0)$ and $(1, 1, 1)$. These are our two target sectors. The three integrals are organised in terms of the t, r and s as follows,

$$\begin{array}{c|ccc} & t & r & s \\ \hline I(1,-1,0) & 1 & 0 & 1 \\ I(1,1,1) & 3 & 0 & 0 \\ I(3,0,0) & 1 & 2 & 0 \end{array} \tag{2.98}$$

then, by order of complexity, the algorithm orders the integrals as,

$$I(1, 1, 1) \succ I(3, 0, 0) \succ I(1, -1, 0). \tag{2.99}$$

After determining the integral families and sectors present, we apply the IBP identities to the seed integrals. These seed integrals are a part of a finite, bounded set of integrals chosen from each relevant sector, whose IBP identities must be sufficiently numerous to reduce every target integral and every non-zero subtopology reached along the way.

After generating IBP identities for the selected seed integrals, any relation containing these representatives is first solved for $I(1, 1, 1)$, the highest-ranked member, and the resulting expression is substituted recursively into the remaining system [28].

IBP reduction may produce lower sectors, but may also generate new integrals from the initially relevant sectors. We may define the relevant sectors as the target sectors plus all lower sectors produced by IBP reduction. Lastly, some sectors may be cancelled due to symmetry, if one denominator is shifted to another under loop-momentum shift or a relabelling of external momenta. If two sectors become equivalent, one is discarded from the independent reduction and the other is retained as its representative. In short, a sector is deemed relevant if it is an original target sector, or a lower sector reached from another via IBP reduction. The relevant sector must also be non-scaleless in a chosen kinematics, not redundant under a family symmetry and, for a cut family, it must retain all required cut denominators throughout.

It is important to note that cut families add a physical restriction: if an IBP reduction step cancels a cut propagator, the sector it produces is missing one of the required on-shell final-state particles and it no longer represents the intended phase-space. If that happens, the integral is set to zero [23].

The reduction is complete when the remaining non-zero family members cannot be eliminated by the available identities. These survivors form a master-integral basis.

2.4.3 Master Integrals

The IBP reduction method reduces the integral to a linear combination of master integrals as,

$$I(\vec{a}) = \sum_{k=1}^{N_{\rm MI}} c_k(\vec{a}, \epsilon, \{s_{ij}\}) M_k \tag{2.100}$$

where c_k are the reduction coefficients and represent what comes from the reduction algebra and M_k are the MI. Our remaining task is evaluating the finite set M_k [28].

Master integrals are not uniquely defined: an integral is a master only relative to the chosen family, kinematics, IBP system, and basis. This means that a different basis with different masters may be used to reduce the same integrals and span the exact same integral space [28].

Master integrals are scalar integrals with a fixed denominator/cut structure. They are generically represented as,

$$M_k = \int \left[\prod_r d^d \ell_r\right] \left[\prod_u d^d p_u\right] \frac{1}{\prod_\alpha D_{v,\alpha}^{a_\alpha} \prod_\beta D_{c,\beta}^{a_\beta}}. \tag{2.101}$$

For integrated antennae, the cut denominators in these master integrals encode the real unresolved phase space. Hence, a master may be a pure phase-space volume, a phase-space integral with invariant denominators, or a loop-plus-phase-space cut integral. The MI are also not necessarily finite, in fact, in dimensional regularisation they commonly contain ϵ poles [23].

For a massless fully inclusive antenna system, q^2 is often the only independent kinematic scale once the dimensional-regularisation normalisation factors have been separated. Through dimensional analysis, the MI are strongly constrained to take the form,

$$M_k(q^2, \epsilon) = \left(q^2\right)^{\lambda_k + \kappa_k \epsilon} f_k(\epsilon) \tag{2.102}$$

where λ_k is fixed by mass dimension and f_k is dimensionless, depending only on ϵ. For this reason, phase-space MI can often be written in terms of Gamma functions, as [23],

$$f_k(\epsilon) \sim \frac{\Gamma^m(1 - a\epsilon)}{\Gamma^n(1 - b\epsilon)}. \tag{2.103}$$

When evaluating master integrals, different types often call for different approaches. Simple one-scale phase-space masters often integrate directly after the phase-space is parametrised. Angular and energy-fraction integrations typically reduce to Beta functions. When a master depends on a non-trivial kinematic variable, such as $x = m^2/q^2$, the most common approach is to differentiate the master with respect to that variable. The derivative produces integrals in the same family, which IBP is then able to reduce to the master basis, as [29],

$$\frac{d}{dx}\vec{M}(x, \epsilon) = A(x, \epsilon)\vec{M}(x, \epsilon). \tag{2.104}$$

where $\vec{M} = (M_1, \ldots, M_{N_{\rm MI}})^{\rm T}$ is the vector of masters and A is a matrix of coefficient functions. Boundary conditions can then be used to determine the solution. These methods are discussed in further detail in Appendix [ref to appropriate appendix].

Once the MI are evaluated, we expand them around $\epsilon = 0$, as a Laurent series [23],

$$M_k = \sum_{n=n_{\min}}^{\infty} m_{k,n}\epsilon^n \tag{2.105}$$

to expose their pole and finite structure. The lowest point $n_{\min}$ can, and often is, negative, resulting in an expansion similar to,

$$M_k = \frac{m_{k,-2}}{\epsilon^2} + \frac{m_{k,-1}}{\epsilon} + m_{k,0} + O(\epsilon). \tag{2.106}$$

The required expansion depth is determined by the reduction coefficient. If $c_k \sim \epsilon^{-p}$, the M_k must be expanded through to ϵ^p to obtain the finite contribution to $c_k M_k$.

2.5 The Massive Extension

Up until now every part of the theory has been explained generally, but with a clear preference for the massless regime. That is because the vast majority of the present work's ensuing chapters will focus on that regime. But a still significant part of it will explore an extension of the discussed methods into the massive regime.

The fundamental change is the massive on-shell condition; the remainder of this section traces its consequences for the antenna construction and integration. Since final-state particles now have mass, their squared momenta are not set to zero, they are set to their squared mass.

$$p_q^2 = p_{\bar{q}}^2 = 0 \xrightarrow{\text{Massive Regime}} p_q^2 = p_{\bar{q}}^2 = m_q^2. \tag{2.107}$$

where p_i represents the momentum and m_i represents the mass of particle i.

The antenna subtraction formalism and the reduction architecture that was developed and described throughout this chapter are retained, however. What changes is the intermediate steps and kinematics: the massive on-shell condition changes the kinematics themselves and the unresolved limits, with soft radiation remaining singular but heavy-quark collinear behaviour becoming quasi-collinear. Massive integration is also different as the massive phase-scale is not equal to its massless counterpart. A new scale of m_q^2/q^2 is also introduced and leads to distinct cut-integral families and master integrals [30].

2.5.1 Massive Kinematics and Infrared Structure

In Subsection 2.1.1, the Mandelstam invariants were introduced for the massless regime. The general definition is,

$$\begin{aligned} s_{ij} = (p_i + p_j)^2 &= p_i^2 + 2p_i \cdot p_j + p_j^2 \\ &= m_i^2 + m_j^2 + 2p_i \cdot p_j. \end{aligned} \tag{2.108}$$

Through momentum conservation, which states that $q = \sum_i p_i$, we also have to reconsider the q^2 expression, as,

$$q^2 = \sum_i m_i^2 + 2\sum_{i<j} p_i \cdot p_j. \tag{2.109}$$

Soft radiation remains singular in the massive regime and therefore continues to generate explicit infrared poles after integration. A non-zero mass regulates collinear limits involving a massive parton, whereas collinear limits between two massless partons remain singular. If the two final-state collinear partons are massless, then the limit follows the massless collinear limit from the massless regime. If the two partons are massive, no strict collinear singularity exists. But if one of the collinear partons is massive, consider that particle i of mass m_i and energy E_i, and the other is massless, consider that particle j, such that $m_j = 0$, then,

$$p_i \cdot p_j = E_i E_j (1 - \beta_i \cos\theta), \quad \beta_i = \sqrt{1 - \frac{m_i^2}{E_i^2}} < 1. \tag{2.110}$$

As the angle between the partons θ tends towards zero, the mass prevents the invariant from reaching the collinear singular surface,

$$\lim_{\theta\to 0} p_i \cdot p_j = E_i E_j (1 - \beta_i) \neq 0. \tag{2.111}$$

This regulated collinear region still carries physically important information, particularly in the joint high-energy and small-angle region, $E_i >> m_i$ and $\theta << 1$,

$$p_i \cdot p_j \sim \frac{E_i E_j}{2} \left(\theta^2 + \frac{m_i^2}{E_i^2}\right) \tag{2.112}$$

and consequently since,

$$\int_0^{\theta^2_{\max}} d\theta^2 \, \frac{1}{p_i \cdot p_j} \sim \int_0^{\theta^2_{\max}} d\theta^2 \left(\theta^2 + \frac{m_i^2}{E_i^2}\right)^{-1} \sim \ln\left(\frac{E_i^2}{m_i^2}\right) \tag{2.113}$$

the massive-massless quasi-collinear region replaces the strict collinear pole by a mass logarithm. The quasi-collinear limit, then, is the limit in which both the mass m_i and the opening angle θ become small together. It is through this limit that the massive antennae recover the massless splitting behaviour, as $m_i \to 0$ [30].

Mass does not regulate UV divergences, as $\ell^2 >> m_i^2, p_i^2$. It is noteworthy, however, that in a theory with massive quarks, the quark mass and quark field must be renormalised as,

$$g_{s,0} = \mu^\epsilon Z_g g_s, \quad m_{q,0} = Z_m m_q, \quad \psi_{q,0} = \sqrt{Z_2^q}\psi_q. \tag{2.114}$$

2.5.2 Massive Phase-Space Integration and Reduction Methods

For final-state particles of momenta p_i and masses m_i, the phase-space becomes,

$$d\Phi_n(q; p_1, ..., p_n) = (2\pi)^d \delta^{(d)}\left(q - \sum_{i=1}^n p_i\right) \prod_{i=1}^n \frac{d^d p_i}{(2\pi)^{d-1}} \delta^+\left(p_i^2 - m_i^2\right). \tag{2.115}$$

At the level of the phase-space measure, the formal change from the massless case, as described in Subsubsection 2.4.2, is,

$$\delta^+(p_i^2) \;\to\; \delta^+(p_i^2 - m_i^2) \tag{2.116}$$

which physically imposes that $p_i^2 = m_i^2$ and $p_i^0 > 0$.

Phase-space factorisation still holds for the massive regime, but the momentum mapping must preserve the relevant mass-shell conditions. So,

$$d\Phi_{m+r}(\{p\}; q) = d\Phi_m(\{\tilde{p}\}; q) d\Phi_X. \tag{2.117}$$

For a final-final antenna with hard radiator i, k and unresolved particles between them, this is schematically,

$$d\Phi_{m+r} = d\Phi_m(..., \tilde{p}_I, \tilde{p}_K, ...; q) d\Phi_{X_{ijk}} \tag{2.118}$$

while $\tilde{p}_I^2 = m_i^2$, $\tilde{p}_K^2 = m_k^2$ and momentum conservation is satisfied,

$$\tilde{p}_I + \tilde{p}_K = p_i + \left(\sum_{\text{unresolved } j} p_j \right) + p_k. \tag{2.119}$$

Antenna integration is computed in the same manner as in the massless regime,

$$\mathcal{X}_n^l \sim \int d\Phi_X \left(\prod_{r=1}^{l} d^d \ell_r \right) X_n^l(\{p\}, \{\ell\}) \sim \left(q^2\right)^{\lambda + \kappa\epsilon} f\left(\left\{ \frac{m^2}{q^2} \right\}, \epsilon \right). \tag{2.120}$$

PaVe and IBP reduction remain applicable in the massive regime, following the same principles introduced for the massless case. For IBP reduction, however, the massive case is significantly harder analytically: the phase-space in non-trivial and has mass-dependent thresholds; the master integrals are generally multi-scale, and can involve logarithms, polylogarithms and more complicated transcendental functions; and lastly, boundary conditions are less direct, although common choices are the massless limit at $m \to 0$ and the threshold limit at $\beta \to 0$ [30].

Chapter 3

Package Framework

AntCalc is an active research *Wolfram Language* package developed to offer direct access to QCD antenna functions, both before and after integration, by automating the symbolic steps required to build and integrate the supported antenna functions.

To complete its function, *AntCalc* makes use of the *FeynCalc* environment for symbolic-amplitude work, and *LiteRed2* for IBP-backed routes. It uses a profile-based system to enable modularity. Its internal architecture is organised as a profile-driven route system: route-specific information is centralised in profiles, which guide the selection and configuration of the appropriate build, reduction, and integration workflows. This establishes a common contract between the package components, reducing the amount of route-specific logic that must be duplicated when existing functionality is extended or new routes are introduced.

The package supports public and experimental routes. The public antenna building and integration routes, which for the current version compute all massless quark-antiquark final-final antenna functions up to NNLO, have been validated against the literature. Experimental routes are unfinished or unchecked and should not be considered validated.

AntCalc's inner working structure can be described using the diagram in Figure 3.1.

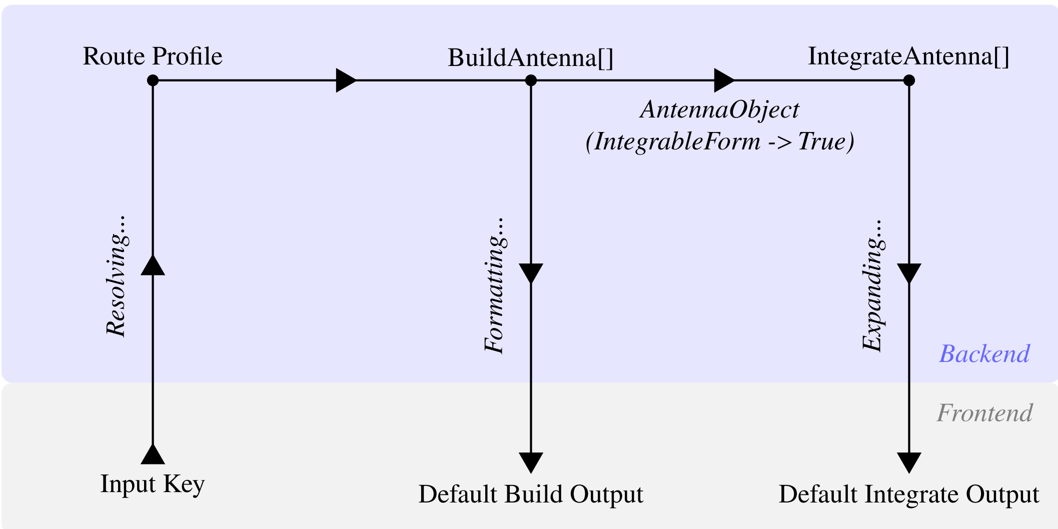

Figure 3.1: Profile-driven architecture of *AntCalc*. Route profiles configure the build and integration workflows, while the build-side route context carries the evolving calculation data. The `AntennaObject` provides the integration-ready handoff between the two stages.

3.1 Design Philosophy and Scope

AntCalc was built to combine the symbolic amplitude capabilities of the FeynCalc environment with the IBP-reduction capabilities of LiteRed2, adapting and orchestrating the parts of those systems required for the construction and integration of QCD antenna functions.

The central design objective of the package is to return results in the conventional forms used in antenna calculations. Though it supports a wide variety of options to inspect and configure the backend operations, the default outputs of its two central functions are designed to reflect this objective. For example, for loop antennae, the default `BuildAntenna[...]` result is given in PaVe-reduced form, as is customary in the literature. The default `IntegrateAntenna[...]` result is also a Laurent ϵ-series up[] to ϵ^0, matching the conventional truncation order, though higher order terms are accessible through the use of the aforementioned options.

Its structure was also designed to be modular and extensible. Each of the two central functions (`BuildAntenna` and `IntegrateAntenna`) is implemented as an encapsulated public module, which communicates with the rest of the codebase using the profile system: the route profile selects and configures the appropriate workflows and the route context carries the evolving results and diagnostics between stages. Using route profiles and route contexts, as described in Section 3.2, the addition of further modules is made easier, reducing duplicated logic and direct dependencies. These new modules need not be compatible with previous implemented functions through function-specific couplings. They need only conform with the profile system, being selected and configured by route profiles, and accepting inputs from and delivering outputs to the route contexts.

Currently, *AntCalc* is organised into two tiers: public and experimental. It already has all massless quark-antiquark final-final antenna functions up to NNLO implemented as part of the public tier, along with every other object required to compute the massless R-ratio up to that perturbative order. These are:

- A_2^0;
- A_2^1;
- A_2^2, $\tilde{A}_2^2$, $\hat{A}_2^2$, $\check{A}_2^2$;
- A_3^0;
- A_3^1, $\tilde{A}_3^1$, $\hat{A}_3^1$;
- A_4^0, $\tilde{A}_4^0$;
- B_4^0;
- C_4^0.

The massive A_3^0 route is also accessible, but remains experimental.

3.2 Profile-Driven Architecture

As described in Section 3.1, *AntCalc* relies on a profile-driven routing system. This system maps the route key to its route-specific profile and workflow.

A public antenna route key is a three-element list which describes a canonical route whose execution may be refined by options. The route key takes the form,

$$\{\text{Antenna Type}, \text{Final-State Multiplicity}, \text{Loop Order}\}. \tag{3.1}$$

As an example, the route key for the one-loop three-parton [1] antenna set is $\{\mathrm{A}, 3, 1\}$, and represents not only the leading A_3^1 antenna, but also its subleading and quark-loop counterparts, and its default public output is the ordered list $\{A_3^1, \tilde{A}_3^1, \hat{A}_3^1\}$, in this specific order.

The package links this key to the profile it identifies and follows that profile's directions on how the route should be constructed and/or integrated. The data produced while the route executes is also stored in the route context association. The profile system does not consist of a single profile, but rather of three main and some conventional auxiliary profiles which are routinely read while the route executes. The principal build-side profile is `AntennaProfile[key]`, which specifies information such as antenna type, multiplicity and loop order, colour normalisation, extraction rules, components, etc. `AntennaReductionProfile[key]` is separate, but still build-side. It specifies the PaVe-reduction preferences for the appropriate routes. In this case, up to NNLO, these are the $\{\mathrm{A}, 2, 1\}$ and $\{\mathrm{A}, 3, 1\}$ loop routes. `AntennaIntegrationProfile[key]` is the integration-side profile, which specifies its side's backend, IBP basis families, expansion depth, etc. The specifics of each stage's profiles and options will be discussed in their respective sections, 3.3 and 3.5.

While profiles select and configure computation paths, the workflows themselves perform the calculations. The route dispatcher and workflow code consult the profiles to select and configure each stage. As an example, which will be further explained and explored in the two upcoming sections, let us consider the call `BuildAntenna[A,2,1]`. Its full route profile is,

```
AntennaProfile[{A, 2, 1}] :=
    <|
        "Key" -> {A, 2, 1},
        "Name" -> "A21",
        "AntennaType" -> A,
        "NumFinalParticles" -> 2,
        "LoopOrder" -> 1,
        "TreeAmplitude" -> AntennaAmplitude[{A, 2, 0}],
        "BornInterference" -> BornInterference[],
        "Extraction" -> "LoopScalar",
        "ColourNorm" -> colourNorm,
        "ExcludedParticles" ->
            {S[_], V[1], V[2], V[3], U[1], U[2], U[3], U[4]},
        "DropSumOver" -> True,
        "ReductionProfile" -> AntennaReductionProfile[{A, 2, 1}],
        "ConventionProfile" -> BuildAntennaConventionProfile[{A, 2, 1}]
    |>
```

In this profile, the key, name, antenna type, number of final particles (multiplicity), and loop order have been discussed before. The remaining entries configure the construction of the route: `TreeAmplitude` supplies the tree-level A_2^0 amplitude with which the one-loop amplitude is interfered, `BornInterference`

[1]It is understood that throughout the rest of the present work, unless explicitly mentioned, all antenna functions considered are massless quark-antiquark final-final antennae.

selects the Born-level self-interference used to normalise the loop interference. `Extraction` selects the scalar one-loop extraction appropriate for this antenna, while `ColourNorm` provides the colour normalisation used. `ExcludedParticles` and `DropSumOver` are settings passed to the *FeynCalc*/*FeynArts* part of the build workflow, selecting what particles are not to be considered in the Feynman diagram generation step and requesting the removal of external `SumOver` structures. Lastly, `ReductionProfile` commands the workflow to complete a lookup on the selected route's antenna, A_2^1 in this case, and determines whether it requires the PaVe reduction backend, from within its `AntennaReductionProfile` association. Since it is a one-loop antenna, it does. `ConventionProfile` selects the convention bridge so the public results follow *AntCalc*'s stated normalisation and dimensional-regularisation conventions, through its separately defined `BuildAntennaConventionProfile` association.

3.3 The Build Stage

This first stage is tasked with building the unintegrated antenna functions from their respective Feynman diagrams. Its central function is `BuildAntenna[type, multiplicity, loop order].` The route through this stage begins by taking the route key from some public call, like `BuildAntenna[A, 3, 1].` It then resolves the corresponding build workflow from the route key, and implements it. Finally, it produces a rich build-side, backend, association which is then reformatted into the public unintegrated antenna expression or component list. This workflow is structured as follows,

```
BuildAntenna[type, multiplicity, loop order]
    -> route key;
    -> loop-order dispatcher;
    -> route-specific build workflow;
    -> build-data association / route context;
    -> public-output boundary;
    -> output: expression, component list, AntennaObject, diagnostics,
       or run record.
```

The loop-order dispatcher is the most important element of this workflow. It dispatches the route, using its key, to its loop order's path, as different loop orders necessitate different mechanisms, the most important of which being the introduction of the reduction of the loop momenta ℓ^μ for one- and two-loop antennae.

Functionally, the route-context for this stage usually contains the initial profile itself (the declarative `AntennaProfile`), the amplitude generated from it, the sectors, computed interferences, components after separation, diagnostics, and its normalised interference. Depending on the route, the context might also retain prototype components, full colour components, reference square components or loop-specific amplitudes.

A diagram in the spirit of the one in Fig. 3.1, but only relative to the Build stage is shown below, in Fig. 3.2.

3.3.1 Tree-Level Routes

As mentioned, the workflow handles distinct loop-order routes differently, though the differences only appear after the dispatcher is called. Tree-level routes like those relative to the A_3^0 antenna, start by retrieving the profile and the interference type (`BornInterference` in this case), and computing the

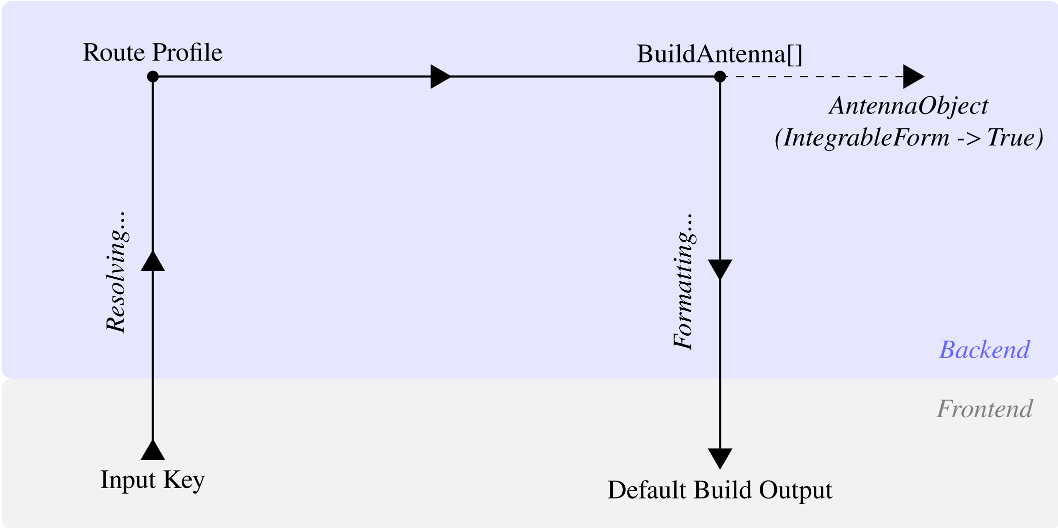

Figure 3.2: Profile-driven architecture of *AntCalc*'s Build stage.

relevant process's amplitude. It uses the profile's `Production` field to select the physical construction path, and the `Extraction` and `ColourNorm` fields to select the extraction mode and the colour normalisation used for each antenna. The colour normalisation used by default has been described previously in Eq.2.24, as $c_N \equiv N - \frac{1}{N}$. The `Extraction` mode also introduces the Born normalisation as described in Eq. 2.19 as $B_0 \equiv |\mathcal{M}_2^0|^2$. The routes selected in the `Production` and `Extraction` fields depend on the methodology required to compute the antenna functions and retrieve the components, if required. The public tree-level antenna functions are constructed using different production and extraction modes, as described in Table 3.1.

Production Mode	Extraction Mode	Current Route
`SelfInterference`	`BornScalar`	A_2^0
`SelfInterference`	`TreeColourCoefficients`	A_3^0
`ColourOrderedAntenna`	`TreeColourCoefficients`	$A_4^0, \tilde{A}_4^0$
`SectorSelfInterference`	`ColourNormScalar`	B_4^0
`SectorSymmetrisedInterference`	`MinusTwoColourNormOverSUNN`	C_4^0

Table 3.1: Overview of the tree-level production and extraction modes.

These production paths all lead to the extraction of the required antenna functions, but the way in which they do it changes with the antenna's multiplicity. `SelfInterference` starts by generating the amplitude, computing its self-interference, and then extracting the antenna. `ColourOrderedAntenna` generates the full-process amplitude and computes the full-colour interference. It then applies the colour-ordered reconstruction as specified by `ColourOrderedSpecs`, and then extracts the components, in this case the Leading and Subleading components. `SectorSelfInterference` generates the full-process amplitude and splits it into profile-defined sectors. It selects the relevant sector, computes its self-interference and extracts the antenna function. Finally, `SectorSymmetrisedInterference` starts by generating the full-process amplitude, and splitting it into direct and exchange sectors. It then computes the required symmetrised interference and extracts the antenna. A reference square is also retained for diagnostics and comparison. The extraction paths lead to the separation of the whole-process results into colour-independent antenna functions. The `BornScalar` mode is used solely in A_2^0, and it does not algebraically divide the computed interference by its Born form, establishing the antenna as the reference normalisation. It returns $A_2^0 = 1$. `TreeColourCoefficients`, as used for A_3^0, A_4^0 and $\tilde{A}_4^0$, starts by normalising the computed amplitude by the colour and the Born normalisation. It then simplifies and expands the normalised interference and searches for the N, $1/N$ and N_f coefficients, the `Leading`, `SubLead` and `QuarkLoop` coefficients in the code. If the terms are present, as described in the previous text, the interference $\mathcal{J}$ becomes,

$$\frac{\mathcal{J}}{c_N B_0} = N X_{\text{Lead}} + \frac{1}{N} X_{\text{SubLead}} + N_f X_{\text{QuarkLoop}} \tag{3.2}$$

and the X terms extracted as the antenna functions. If there are no terms of that shape, like in the A_3^0 case, then $X = \mathcal{J}/(c_N B_0)$, colour- and Born-normalised. It is important to note that, though present, no quark loop antenna should ever appear in the tree-level route. The dispatcher still considers it nonetheless since it is written to be reusable. For B_4^0, the `ColourNormScalar` mode is used, after the primary-current sector has been self-interfered. The extraction then returns the colour- and Born-normalised sector interference, as,

$$B_4^0 = \frac{\mathcal{J}_{\text{sector}}}{c_N B_0}. \tag{3.3}$$

Lastly, `MinusTwoColourNormOverSUNN` is used for C_4^0, and it takes the symmetrised interference of the direct and exchange sectors and normalises it as,

$$C_4^0 = -\frac{N \mathcal{J}_{\text{sym}}}{2 c_N B_0}. \tag{3.4}$$

The C_4^0 antenna requires a further $-N/2$ normalisation factor due to its origin as the interference of the direct and exchange flavour-flow amplitudes for identical quarks. The minus sign reflects the exchange of identical fermions, the $1/2$ factor is required when the symmetrised source contains both cross terms, as discussed in Eq. 2.28. The remaining N factor, along with the usual c_N in the denominator adjusts the interference's colour coefficient into the colour-free C_4^0 antenna.

3.3.2 One-Loop Routes

One-loop routes are dispatched to `BuildLoopAntennaData` where the primary backend reduction method is selected through the nested `ReductionProfile`, which by default is the reduction using PaVe, as described in Subsection 2.4.1.

The production method consists of generating the tree-level and one-loop amplitudes, using `TreeAmplitude` and `MAmpOneLoop`, respectively, and then forming the tree-loop interference. The interference is colour- and one-loop-normalised, using c_N and $\Lambda_l = (8\pi^2)^l$, for loop order l, as described in Eq. 2.9. To this normalised interference is then applied the reduction and the components extracted. These antenna functions also differ in production and extraction method, however. Their production and extraction modes are shown in Table 3.2, below.

Production Mode	Extraction Mode	Current Route
`TreeAmplitude + MAmpOneLoop`	`LoopScalar`	A_2^1
`TreeAmplitude + MAmpOneLoop`	`LoopColourCoefficients`	A_3^1, $\tilde{A}_3^1$, $\hat{A}_3^1$

Table 3.2: Overview of the one-loop production and extraction modes.

Both antenna sets are produced in the same manner, as discussed. The relevant tree-level antenna amplitude, $\mathcal{M}_2^0$ and $\mathcal{M}_3^0$ respectively for A_2^1 and the A_3^1 set, is first computed. Then it is interfered with the loop amplitudes, $\mathcal{M}_2^1$ and $\mathcal{M}_3^1$. The interference is computed, before being normalised as,

$$\mathcal{J}_n^l = 2\,\mathrm{Re}\left(\left(\mathcal{M}_n^0\right)^\dagger \mathcal{M}_n^l\right) \tag{3.5}$$

where n represents the number of final-state particles, as usual.

Since A_2^1's interference does not decompose into colour components, $A_2^1 = \Lambda_1 \mathcal{J}_2^{1\prime} = \Lambda_1 \mathcal{J}_2^1/(c_N B_0)$. The A_3^1 set, on the other hand, appears closely like the A_4^0 set, but with three, rather than two components, the leading, subleading and quark loop antenna functions. The interference is given as in Eq. 3.2. Lastly, as discussed in Eq. 2.13, together with Eq. 2.19, the A_3^1 set needs to be UV renormalised. Consider the notation $X_n^{l,\text{bare}}$ here to denote a bare antenna, such that, for example, $A_2^{1,\text{bare}} = A_2^1$. For the A_3^1 set, the interference expands to,

$$\frac{\Lambda_1}{c_N B_0}\mathcal{J}_3^1 = N T_{\text{Lead}} + \frac{1}{N} T_{\text{Sublead}} + N_f T_{\text{QL}}. \tag{3.6}$$

Differently from other processes at this level, the A_3^1 route necessitates a further subtraction, as the interference components come in terms of a genuine one-loop three-parton antenna and an iterated lower-order contribution. That makes it so that the bare (before UV renormalisation) A_3^1 set antenna functions become,

$$\begin{aligned} A_3^{1,\text{bare}} &= T_{\text{Lead}} - A_2^1 A_3^0 \\ \tilde{A}_3^{1,\text{bare}} &= -\left(T_{\text{Sublead}} + A_2^1 A_3^0\right) \\ \hat{A}_3^{1,\text{bare}} &= T_{\text{QL}}. \end{aligned} \tag{3.7}$$

The components are then, finally, renormalised as discussed,

$$\begin{aligned} A_3^1 &= A_3^{1,\text{bare}} - \frac{11}{6\epsilon} A_3^0 \\ \tilde{A}_3^1 &= \tilde{A}_3^{1,\text{bare}} \\ \hat{A}_3^1 &= \hat{A}_3^{1,\text{bare}} + \frac{1}{3\epsilon} A_3^0. \end{aligned} \tag{3.8}$$

3.3.3 Two-Loop Routes

Loop order two routes are dispatched to `BuildTwoLoopAntennaData`. There is only, formally, one set of this kind, A_2^2. It is remarkably different from the rest, however. First, these objects, alongside A_2^0 and A_2^1, are not formally antenna functions as they fail to meet the definition of a structure that captures the singular behaviour of an unresolved parton radiated between two hard, colour-connected partons [11]. They are, instead, form factors required by the antenna framework. The A_2^2 set, at two loops, is obtained from the two-loop quark form factor [24]. Second, this four-component set is not generated like A_4^0 and A_3^1 were, rather these four components come from two distinct source contributions. Their production and extraction modes are similar and handled as a bundle inside *AntCalc*, but, as shown in Table 3.3, their sources differ,

The first three antenna objects are formed by interfering the LO tree-level amplitude with the two-loop

Contribution	Extraction Mode	Current Route
`TwoLoopInterf`	`TwoLoopTTermComponents`	A_2^2, $\tilde{A}_2^2$, $\hat{A}_2^2$
`OneLoopSelf`	`TwoLoopTTermComponents`	$\breve{A}_2^2$

Table 3.3: Overview of the two-loop contributions and extraction modes.

amplitude. This interference follows the blueprint defined in Eq. 3.5, as the interference becomes,

$$\mathcal{J}_2^{2,[2\times 0]} = 2\,\mathrm{Re}\left(\left(\mathcal{M}_2^0\right)^\dagger \mathcal{M}_2^2\right). \tag{3.9}$$

The superscript $[2\times 0]$ denotes the nature of the operation: interfering a two-loop with a zero-loop amplitude. It is also usual to split the T-object relative to these four components, $T_{q\bar{q}}^6$, into two, based on the two contributions.

The last object, $\breve{A}_2^2$, comes from self-interfering the one-loop amplitude as,

$$\mathcal{J}_2^{2,[1\times 1]} = \left(\mathcal{M}_2^1\right)^\dagger \mathcal{M}_2^1. \tag{3.10}$$

Both of these interferences are equally Born- and loop-normalised, using B_0 and Λ_2. They are also colour-normalised, but while $\mathcal{J}_2^{2,[2\times 0]}$ requires only c_N, $\mathcal{J}_2^{2,[1\times 1]}$ requires c_N^2 since it is computed using two one-loop amplitudes, each of which requires a c_N normalisation factor, as described in Subsection 2.3.2. They, then, become,

$$\begin{aligned}\frac{\Lambda_2}{c_N B_0}\mathcal{J}_2^{2,[2\times 0]} &= NA_2^{2,\mathrm{bare}} + \frac{1}{N}\tilde{A}_2^{2,\mathrm{bare}} + N_f\hat{A}_2^{2,\mathrm{bare}} \\ \frac{\Lambda_2}{c_N^2 B_0}\mathcal{J}_2^{2,[1\times 1]} &= \breve{A}_2^{2,\mathrm{bare}}.\end{aligned} \tag{3.11}$$

The antenna objects have to then be UV renormalised as follows,

$$\begin{aligned} A_2^2 &= A_2^{2,\mathrm{bare}} - \frac{11}{6\epsilon}A_2^1 \\ \tilde{A}_2^2 &= \tilde{A}_2^{2,\mathrm{bare}} \\ \hat{A}_2^2 &= \hat{A}_2^{2,\mathrm{bare}} + \frac{1}{3\epsilon}A_2^1 \\ \breve{A}_2^2 &= \breve{A}_2^{2,\mathrm{bare}}.\end{aligned} \tag{3.12}$$

These renormalised results are now used to form the overall $T_{q\bar{q}}^6$ object. It is defined as the sum of the two contributions' sub-T-objects, $T_{q\bar{q}}^{6,[2\times 0]}$ for the first, and $T_{q\bar{q}}^{6,[1\times 1]}$ for the second contribution, following the interferences $\mathcal{J}$ notation as,

$$\begin{aligned} T_{q\bar{q}}^6 &= \left(N - \frac{1}{N}\right) T_{q\bar{q}}^2 \left[T_{q\bar{q}}^{6,[2\times 0]} + T_{q\bar{q}}^{6,[1\times 1]}\right] \\ &= \left(N - \frac{1}{N}\right) T_{q\bar{q}}^2 \left[\frac{\Lambda_2}{c_N B_0}\mathcal{J}_2^{2,[2\times 0]} + \frac{\Lambda_2}{c_N B_0}\mathcal{J}_2^{2,[1\times 1]}\right] \\ &= \left(N - \frac{1}{N}\right) T_{q\bar{q}}^2 \left[NA_2^2 + \frac{1}{N}\tilde{A}_2^2 + N_f\hat{A}_2^2 + \left(N - \frac{1}{N}\right)\breve{A}_2^2\right].\end{aligned} \tag{3.13}$$

3.4 Public Boundary and Integration handoff

As previously discussed, each `BuildAntenna` workflow produces a route context. That context contains the initial profile alongside some selected pieces of data acquired during the building process. The ordinary, default, `BuildAntenna[...]` call formats this contexts into the conventional public output, which consists of a single expression or ordered list of components. This output is not compatible with a following `IntegrateAntenna[...]` call, as this second call requires the route profile information. To make the two functions compatible we must use the handoff option `IntegrableForm`. Calling `BuildAntenna[..., IntegrableForm -> True]` returns an `AntennaObject` instead of the default public output. It retains the route key, the selected output, the profile, the complete build data and the source-contribution provenance. This `AntennaObject` is the intended input of the `IntegrateAntenna[...]` function.

3.5 The Integration Stage

The integration stage is tasked with taking the complete `AntennaObject`, as produced by a `BuildAntenna[..., IntegrableForm -> True]`, resolving the integration route from the key, performing all intermediate calculations and applying all necessary conventions, and finally returning a Laurent-expanded integrated antenna or component list. The workflow's structure is,

```
IntegrateAntenna[AntennaObject]
    -> recover route key and resolve profile;
    -> choose IBP backend;
    -> complete IBP reduction;
    -> substitute master integrals when applicable;
    -> extract public integrated antennae;
    -> output: selected component, list, diagnostics, run record, or
       master-combination view.
```

It is relevant to note that `IntegrateAntenna` requires the profile to complete its function like `BuildAntenna`, since it also encodes the relevant routing information for the reduction and integration process.

Much like for `BuildAntenna`, there is no single route context object, rather the relevant information is stored around multiple objects that are accessed when required. These objects store initiation-crucial elements, like the key and the profile, the selected component and the input (unintegrated) antenna, some IBP-crucial elements, like the bases required. the master integrals to use, and the series result, along with others like the route story, or the diagnostics.

An integration-stage-specific diagram in the spirit of the one in Fig. 3.1 can also be drawn in further detail. That diagram is shown in Fig. 3.3.

3.5.1 Integration Profiles and Selection

The integration-specific profile for each route selects the kind of backend process (and external package workflow used), and the basis family required for the process. The antennae are divided by topology, that is, number of final-state particles and number of loops, and all of those that share the same topology in the current implementation, share the same basis-family set. The exception to this rule is the

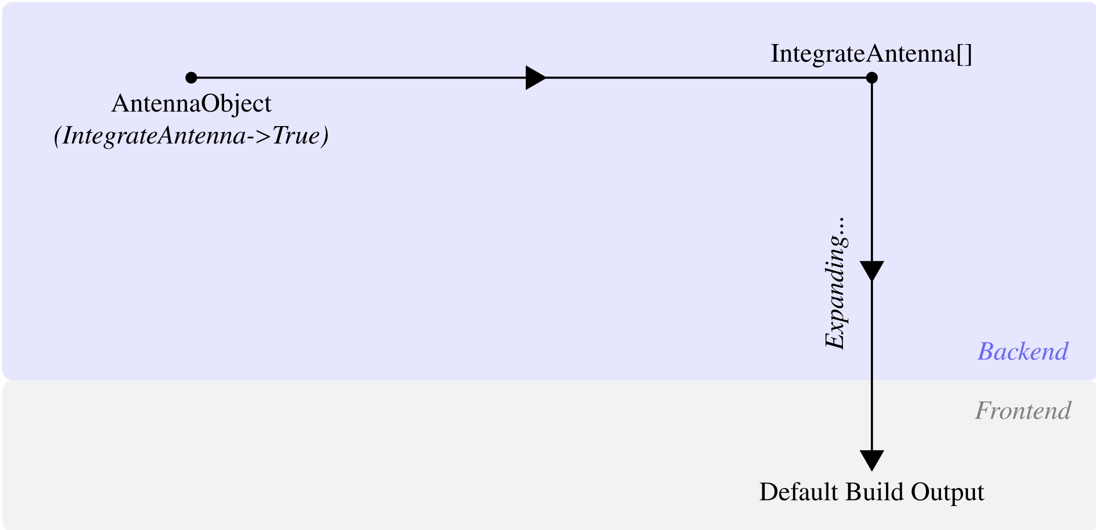

Figure 3.3: Profile-driven architecture of *AntCalc*'s Integration stage.

A_2^2 set, in which different sets are used depending on configuration. The A_2^1 antenna is also an exception to the use of basis families, as it is reduced using PaVe reduction and then the PaVe MI are substituted by their analytical results in terms of ϵ. An overview of these is shown in the following Table 3.4.

Default Backend	Bases-Family Set	Current Route
PaVe / *FeynHelpers*	massless two-parton vertex	A_2^1
IBP / *LiteRed2*	`X30`	A_3^0
IBP / *LiteRed2*	`X40`	A_4^0, $\tilde{A}_4^0$, B_4^0, C_4^0
IBP / *LiteRed2*	`A31`	A_3^1, $\tilde{A}_3^1$, $\hat{A}_3^1$
IBP / *LiteRed2*	`A22TwoLoopTree`	A_2^2, $\tilde{A}_2^2$, $\hat{A}_2^2$
IBP / *LiteRed2*	`A22OneLoopSelf`	$\check{A}_2^2$

Table 3.4: Overview of the integration-side route-profiles.

As discussed previously for the build-side, the profile here also does not complete any calculations itself. It simply provides the workflow with the required information for it to work. For example, consider the integration of the B_4^0 antenna: it reaches `IntegrateAntenna` as an `AntennaObject`. For the key, the workflow is able to resolve the integration-side route-profile and then it reduces the input unintegrated antenna from `AntennaObject` with the IBP method, using *LiteRed2*, and using the bases and MI from the `X40` basis-family set.

3.5.2 Integration Routes

PaVe Route

The A_2^1 PaVe route is the distinct low-multiplicity virtual case. It is the only route that does not involve IBP. Due to $A_2^0 = 1$, this A_2^1 antenna's integration workflow requires only the reduction and integration of the single loop momentum, making PaVe reduction the best choice. The workflow then uses *Package-X*, which comes pre-installed within *FeynHelpers*, to evaluate the PaVe scalar one-loop functions and, after applying the required branch and scale conventions, the kinematic-scale normalisation, and Laurent expansion in ϵ, it gives the integrated $\mathcal{A}_2^1$ antenna. This process is shown schematically in Fig. 3.4, below.

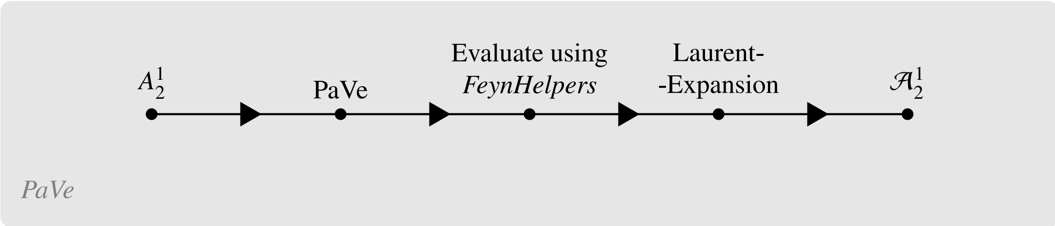

Figure 3.4: PaVe reduction for the A_2^1 antenna integration.

IBP Route

All other antenna functions are integrated using IBP reduction using *LiteRed2* and as described in Sections 2.4. The workflow rewrites the momenta, invariants, and denominators in *LiteRed2*'s terms. It then adds the phase-space cut structure for the selected family, loads its bases and after expanding the unintegrated antenna expression, going term-by-term, matches each one to its relevant family. After each match, the term is IBP reduced relative to its matched family-basis MI. This last step is handled completely by *LiteRed2*'s internal engine. After all terms have been reduced, they are summed to form the final reduced antenna function. For the public routes, the MI have already been hand-matched with known MI. These known MI are then substituted by their values after integration, which is also hand-calculated and finally, after applying all required normalisation and conventions, the integrated antenna functions are Laurent-expanded in ϵ. The IBP route process is shown schematically in Fig. 3.5.

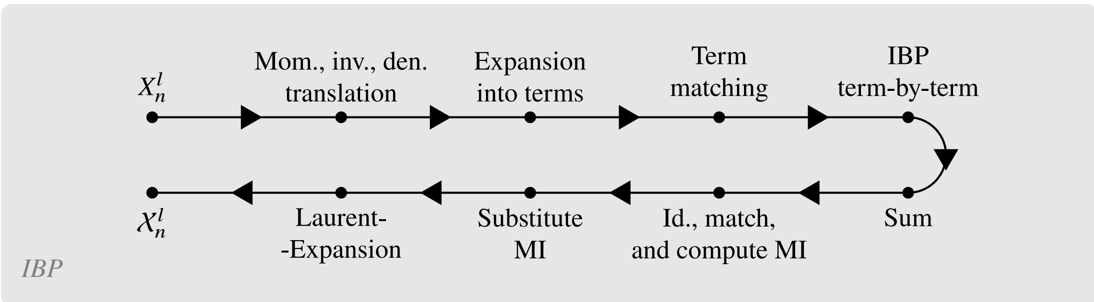

Figure 3.5: IBP reduction workflow for the remaining antennae integrations.

3.5.3 Normalisations and Convention Bridge

As discussed in Section 2.2, before integration, the antenna functions must be normalised using the $\mathcal{N}(\epsilon, k)$ factor, introduced in Eq. 2.12. Since in Section 3.3, where we discussed the Build stage, the unintegrated antenna functions were normalised using the loop-order factor, Λ_l, in this stage, the rest of the normalisations and conventions must be applied. That begins with the application of what remains from the G_k factor and then $C(\epsilon, k)$. Finally, we reach the full $\mathcal{N}(\epsilon, k)$ by diving out the twp-particle phase-space measure. This is, schematically,

$$\begin{aligned} \mathcal{X}_n^l &= \mathcal{N}(\epsilon, k) \int d\Phi_{X_{ijk}}\, X_n^l \\ &= \frac{1}{\Phi_2} \left(8\pi^2\right)^{n-2} S_\epsilon \int d\Phi_{X_{ijk}} \left(\Lambda_l X_n^l\right). \end{aligned} \tag{3.14}$$

Note, however, that not all routes require that the rest of the G_k be applied, based on their n. There also those routes that did not require the application of the Λ_l factor in the Build stage. The following

Table 3.5 shows how each antenna or antenna set interacts with G_k.

Route	n, l	G_k **relative to build** Λ_l
A_2^1	$2, 1$	$G_1 = \Lambda_1$ (no extra multiplicity factors needed)
A_3^0	$3, 0$	$G_1 = 8\pi^2\Lambda_0 = 8\pi^2$
A_3^1 set	$3, 1$	$G_2 = 8\pi^2\Lambda_1$
A_4^0 set, B_4^0, C_4^0	$4, 0$	$G_2 = (8\pi^2)^2\Lambda_0 = (8\pi^2)^2$
A_2^2 set	$2, 2$	$G_2 = \Lambda_2$ (no extra multiplicity factors needed)

Table 3.5: Overview of the interactions of the antenna routes with the normalisation factor G_k.

3.5.4 Component Extraction and Integrated $\mathcal{T}$-Objects

The last part of the Build stage, before it generates the output, is component extraction. In that part, the full result of the calculations the workflow went over is separated into colour-dependent terms and the coefficients relative to each of those terms is labelled as one of the components (leading, subleading, etc.). All multi-component antenna routes are reduced and integrated using IBP reduction, which is independent of colour. That makes it sensible to completely separate the components before reducing. This way, not only does the process become much cheaper if only one component is selected, but the reduction are also made cleaner, since different components may require different bases and MI. The following diagram in Fig.3.6, describes schematically this process, using the A_3^1 antenna set as an example.

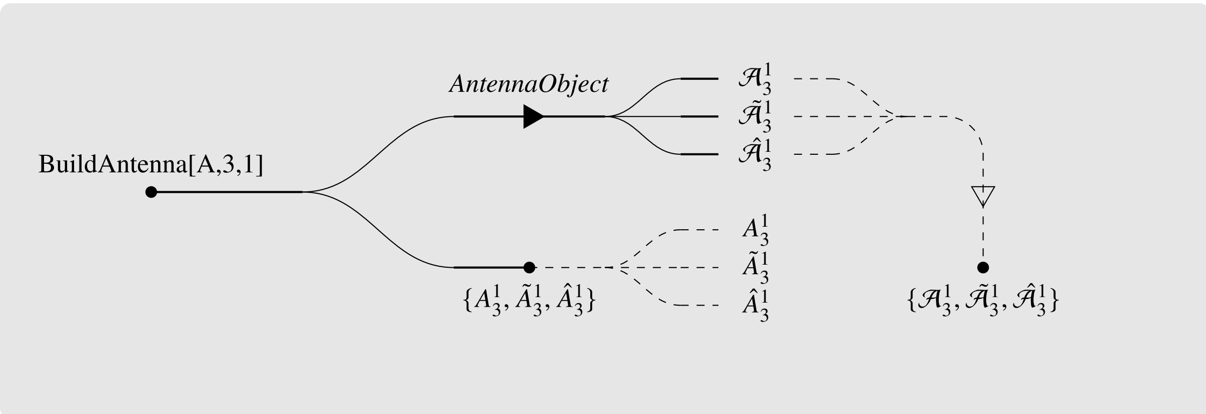

Figure 3.6: Diagram showing the component extraction steps and paths for the build-only and build-and-integrate routes. The combined route is shown above and the build-only route below it. The dashed lines represent the options to either separate the ordered lists into single components on the build-only route, or to merge the components into a single ordered list for the combined route. The A_3^1 antenna set is used as an example.

It is also important to note here that the result of the A_3^1 set's integration is given, much like it was for the Build stage in 3.7, as,

$$\mathcal{A}_3^1 = \mathcal{T}_{\text{Lead}} - \mathcal{A}_2^1\mathcal{A}_3^0$$
$$\tilde{\mathcal{A}}_3^1 = -\left(\mathcal{T}_{\text{Sublead}} + \mathcal{A}_2^1\mathcal{A}_3^0\right).$$

These resulting $\mathcal{A}_3^1$ and $\tilde{\mathcal{A}}_3^1$ are already UV renormalised by this stage.

3.6 Higher-Level Functions

3.7 Validation, Diagnostics, and Provenance

Chapter 4

Worked Example: A_3^0 End To End

4.1 Purpose of the Example

The simplest non-trivial SMQCD antenna is A_3^0. As such, it may be used as an example of how AntCalc handles the physics involved in building and then integrating it over the phase space. In this chapter, we are going to go over how this antenna is mathematically computed by hand, and how AntCalc handles those same calculations on its end.

4.2 Mathematical Derivation

4.2.1 Physics Definition

The A_3^0 is the tree-level final-final quark-antiquark-gluon antenna. It represents the normalised $\gamma^*(p) \to q(k_1)g(k_3)\bar{q}(k_2)$ process. This process, in the current model, is able to produce two distinct topologies which differ by what quark line the gluon is emitted from. Those diagrams are shown below in Figure 4.1.

(a) Gluon emitted from the quark line.

(b) Gluon emitted from the antiquark line.

Figure 4.1: Feynman diagrams depicting the two topologies described under the A_3^0 antenna, corresponding to the process $\gamma^* \to q(k_1)g(k_3)\bar{q}(k_2)$.

4.2.2 Computing the Amplitude

The amplitude is given as follows directly from the diagrams,

$$\mathcal{M}_3^0 = e_q g_s (T^{a_3})_{i_1 i_2} \left[\bar{u}(k_1) \not{\epsilon}^*(k_3) \frac{\not{k}_1 + \not{k}_3}{s_{13}} \not{\epsilon}(p) v(k_2) - \bar{u}(k_1) \not{\epsilon}^*(p) \frac{\not{k}_2 + \not{k}_3}{s_{23}} \not{\epsilon}(k_3) v(k_2) \right] \tag{4.1}$$

where $(T^{a_3})_{i_1 i_2}$ represents the colour matrix for the quark and antiquark in positions 1 and 2 and gluon in position 3.

4.2.3 Squared Matrix Element

The amplitude is now squared as follows,

$$\begin{aligned}
|\mathcal{M}_3^0|^2 &= e_q^2 g_s^2 (T^{a_3})_{i_1 i_2} (T^{a_3})_{i_1 i_2} \left(M_A M_A^\dagger + M_B M_B^\dagger - M_A M_B^\dagger - M_B M_A^\dagger \right) \\
\text{with } M_A &= \bar{u}(k_1) \not{\varepsilon}^*(k_3) \frac{\not{k}_1 + \not{k}_3}{s_{13}} \not{\varepsilon}(p) v(k_2) = \bar{u}(k_1) \Gamma_A v(k_2), \\
M_B &= \bar{u}(k_1) \not{\varepsilon}^*(p) \frac{\not{k}_2 + \not{k}_3}{s_{23}} \not{\varepsilon}(k_3) v(k_2) = \bar{u}(k_1) \Gamma_B v(k_2).
\end{aligned} \tag{4.2}$$

which in total becomes

$$\begin{aligned}
&= \sum_{a_3, i_1, i_2} (T^{a_3})_{i_1 i_2} (T^{a_3})_{i_1 i_2} \\
&\qquad \sum_{s_1, s_2, \lambda_g, \lambda_\gamma} \bar{u}(k_1, s_1) u(k_1, s_1) \left[\Gamma_A \bar{\Gamma}_A + \Gamma_B \bar{\Gamma}_B - \Gamma_A \bar{\Gamma}_B - \Gamma_B \bar{\Gamma}_A \right] v(k_2, s_2) \bar{v}(k_2, s_2)
\end{aligned} \tag{4.3}$$

Each of the Γ terms becomes, using Dirac Algebra [ref to dirac algebra maths],

$$\begin{aligned}
\sum_{s_1, s_2, \lambda_g, \lambda_\gamma} &\bar{u}(k_1, s_1) u(k_1, s_1) \left(\Gamma_A \bar{\Gamma}_A \right) v(k_2, s_2) \bar{v}(k_2, s_2) = \\
&= \frac{1}{s_{13}^2} \sum_{\lambda_g, \lambda_\gamma} \varepsilon_\mu^* \varepsilon_\nu \varepsilon_\rho \varepsilon_\sigma^* \operatorname{Tr} \left(\not{k}_1 \gamma^\mu (\not{k}_1 + \not{k}_3) \gamma^\rho \not{k}_2 \gamma^\sigma (\not{k}_1 + \not{k}_3) \gamma^\nu \right) \\
&= \frac{1}{s_{13}^2} g_{\mu\nu} g_{\rho\sigma} \operatorname{Tr} \left(\not{k}_1 \gamma^\mu (\not{k}_1 + \not{k}_3) \gamma^\rho \not{k}_2 \gamma^\sigma (\not{k}_1 + \not{k}_3) \gamma^\nu \right) \\
&= \frac{1}{s_{13}^2} \operatorname{Tr} \left(\not{k}_1 \gamma^\mu (\not{k}_1 + \not{k}_3) \gamma^\rho \not{k}_2 \gamma_\rho (\not{k}_1 + \not{k}_3) \gamma_\mu \right) \\
&= \left(\frac{2-d}{s_{13}} \right)^2 \operatorname{Tr}(\not{k}_1 \not{P} \not{k}_2 \not{P}), \quad \not{P} = (\not{k}_1 + \not{k}_3) \\
&= 4 \left(\frac{2-d}{s_{13}} \right)^2 \left(2(k_1 \cdot P)(k_2 \cdot P) - (k_1 \cdot k_2) P^2 \right) \\
&= 2(d-2)^2 \frac{s_{23}}{s_{13}}
\end{aligned} \tag{4.4}$$

$$\begin{aligned}
\sum_{s_1, s_2, \lambda_g, \lambda_\gamma} &\bar{u}(k_1, s_1) u(k_1, s_1) \left(\Gamma_B \bar{\Gamma}_B \right) v(k_2, s_2) \bar{v}(k_2, s_2) = \\
&= 2(d-2)^2 \frac{s_{13}}{s_{23}}
\end{aligned} \tag{4.5}$$

$$\sum_{s_1,s_2,\lambda_g,\lambda_\gamma} \bar{u}(k_1,s_1)u(k_1,s_1)\left(\Gamma_A\bar{\Gamma}_B\right)v(k_2,s_2)\bar{v}(k_2,s_2) = \\ = 2(d-2)\frac{(d-4)s_{13}s_{23}-2s_{12}^2-2s_{12}(s_{13}+s_{23})}{s_{13}s_{23}} \tag{4.6}$$

$$\sum_{s_1,s_2,\lambda_g,\lambda_\gamma} \bar{u}(k_1,s_1)u(k_1,s_1)\left(\Gamma_B\bar{\Gamma}_A\right)v(k_2,s_2)\bar{v}(k_2,s_2) = \\ = 2(d-2)\frac{(d-4)s_{13}s_{23}-2s_{12}^2-2s_{12}(s_{13}+s_{23})}{s_{13}s_{23}} \tag{4.7}$$

and computing the colour matrix sum as

$$\sum_{a_3,i_1,i_2}(T^{a_3})_{i_1i_2}(T^{a_3})_{i_1i_2} = \frac{1}{2}(N^2-1) \tag{4.8}$$

we are finally able to reach the final result, as follows, using dimensional regularisation as $d \to 4-2\epsilon$ [ref to dimreg],

$$\begin{aligned}|\mathcal{M}_3^0|^2 &= e_q^2g_s^2(N^2-1)\left(2(d-2)^2\left(\frac{s_{13}}{s_{23}}+\frac{s_{23}}{s_{13}}\right)-4(d-2)\frac{(d-4)s_{13}s_{23}-2s_{12}^2-2s_{12}(s_{13}+s_{23})}{s_{13}s_{23}}\right)\\ &= 2(d-2)e_q^2g_s^2(N^2-1)\left((d-2)\left(\frac{s_{13}}{s_{23}}+\frac{s_{23}}{s_{13}}\right)-2\frac{(d-4)s_{13}s_{23}-2s_{12}^2-2s_{12}(s_{13}+s_{23})}{s_{13}s_{23}}\right)\\ &= \frac{8}{s_{13}s_{23}}\left(2q^2s_{12}+s_{13}^2+s_{23}^2-2\epsilon\left(q^2s_{12}+s_{13}^2+s_{23}^2+s_{13}s_{23}\right)+\epsilon^2\left(s_{13}+s_{23}\right)^2\right)\end{aligned} \tag{4.9}$$

The A_3^0 antenna also requires the $\mathcal{M}_2^0$ amplitude. This amplitude corresponds to the $\gamma^* \to q\bar{q}$ Born process. The amplitude is

$$\mathcal{M}_2^0 = e_q\delta_{i_1i_2}\bar{u}(k_1)\not{\epsilon}(p)v(k_2). \tag{4.10}$$

Applying the same spin, polarization and colour sums as above, and considering that, for a two final-state particles system $s_{12}=q^2$, we get

$$\begin{aligned}|\mathcal{M}_2^0|^2 = 4e_q^2(d-2)Ns_{12} &= 4e_q^2(d-2)Nq^2\\ &= 8e_q^2(1-\epsilon)Nq^2\end{aligned} \tag{4.11}$$

4.2.4 Antenna Building and Normalisation

The antenna is now finally ready to be built. That is accomplished using the following formula,

$$A_3^0 = \frac{1}{2g_s^2C_F}\frac{|\mathcal{M}_3^0|^2}{|\mathcal{M}_2^0|^2}. \tag{4.12}$$

By taking $C_F = (N^2-1)/(2N)$, the formula yields, using Eqs. 4.9 and 4.11,

$$A_3^0 = \frac{2q^2 s_{12} + s_{13}^2 + s_{23}^2}{q^2 s_{13} s_{23}} - \epsilon \frac{(s_{13}+s_{23})^2}{q^2 s_{13} s_{23}}$$
$$= \frac{1}{q^2}\left[\left(\frac{s_{13}}{s_{23}} + \frac{s_{23}}{s_{13}} + 2\frac{s_{12}q^2}{s_{13}s_{23}}\right) - \epsilon\left(2 + \frac{s_{13}}{s_{23}} + \frac{s_{23}}{s_{13}}\right)\right] \tag{4.13}$$

4.2.5 Antenna Integration

We now integrate the A_3^0 antenna over the appropriate (3 final-state particle, in this case) phase-space. This integral is computed in d dimensions and over the phase-space measure in terms of the Mandelstam invariants (s_{13}, q^2, etc.). The integral takes the form,

$$\mathcal{A}_3^0 = C(\epsilon,1)\int d\Phi_{ijk}\, A_3^0$$
$$= \mathcal{N}(\epsilon,1)\,(4\pi)^{3-2d}\left(q^2\right)^{1-\frac{d}{2}}\int ds_{12}ds_{13}ds_{23}d\Omega\,(s_{12}s_{13}s_{23})^{\frac{d-4}{2}}\delta\left(q^2 - s_{12} - s_{13} - s_{23}\right) A_3^0. \tag{4.14}$$

with $d\Phi_{ijk} = d\Phi_3/\Phi_2$ and with the derivation of the $d\Phi_3$ measure being present in [ref to this derivation]. $C(\epsilon,1)$ is the integral normalisation which is here given as

$$C(\epsilon,1) = 8\pi^2(4\pi)^{-\epsilon}e^{\epsilon\gamma_E}. \tag{4.15}$$

This integral is quite complex and spans across all A_3^0 terms. Accordingly, we are not going to compute this integral directly: instead we are going to use IBP reduction to reduce the expression to a linear combination of master integrals, compute those master integrals and then use the result to compute the integrated antenna. This process is explained in [ref to IBP section].

After applying IBP identities and reducing the integral to a linear combination of this master integral[1], we reach the following expression,

$$\mathcal{A}_3^0 = \frac{C(\epsilon,1)}{\Phi_2}\frac{2(2-6\epsilon+5\epsilon^2-2\epsilon^3)}{q^2\epsilon^2}R_3 \tag{4.16}$$

This antenna has a single master: R_3 [ref to master integrals section/appendix]. This master integral is the integral over phase-space itself,

$$R_3 = \int d\Phi_3 = \frac{2^{-7+4\epsilon}\pi^{-3+2\epsilon}\Gamma^3(1-\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)}\left(q^2\right)^{1-2\epsilon}. \tag{4.17}$$

After using the expression in Eq. 4.17 explicitly in Eq. 4.16becomes

$$\mathcal{A}_3^0 = \frac{C(\epsilon,1)}{\Phi_2}\left(q^2\right)^{-2\epsilon}(4\pi)^{-3+2\epsilon}(2-6\epsilon+5\epsilon^2-2\epsilon^3)\frac{\Gamma^3(1-\epsilon)}{\epsilon^2\Gamma(3-3\epsilon)\Gamma(2-2\epsilon)}$$
$$= \left(q^2\right)^{-\epsilon}e^{\epsilon\gamma_E}(1-2\epsilon)(2+\epsilon(\epsilon+2))\frac{\Gamma^2(-\epsilon)}{\Gamma(3-3\epsilon)} \tag{4.18}$$

[1] This process is shown completely in [ref to ibp reduction appendix].

and, after expanding the integrated expression as a Laurent series in ϵ, we finally arrive at the infrared pole structure,

$$\mathcal{A}_3^0 = \left(q^2\right)^{-\epsilon}\left[\frac{1}{\epsilon^2}+\frac{3}{2\epsilon}+\frac{57-7\pi^2}{12}+\epsilon\left(\frac{84-21\pi^2-8\psi^{(2)}(1)+108\psi^{(2)}(3)}{24}\right)\right.$$
$$\left.+\,\epsilon^2\left(\frac{35640-3990\pi^2-71\pi^4-720\psi^{(2)}(1)+9720\psi^{(2)}(3)}{1440}\right)\right]$$

$$= \left(q^2\right)^{-\epsilon}\left[\frac{1}{\epsilon^2}+\frac{3}{2\epsilon}+\frac{19}{4}-\frac{7\pi^2}{12}+\epsilon\left(\frac{109}{8}-\frac{7\pi^2}{8}-\frac{25\zeta_3}{3}\right)+\epsilon^2\left(\frac{639}{16}-\frac{133\pi^2}{48}-\frac{71\pi^4}{1440}-\frac{25\zeta_3}{2}\right)\right] \tag{4.19}$$

where the usual polygamma identities, $\psi^{(2)}(1) = -2\zeta_3$ and $\psi^{(2)}(3) = -2\zeta_3 + \frac{9}{4}$, were employed.

4.3 Computational Derivation

4.3.1 Antenna Building

Using AntCalc, building the antenna is very simple, since the function `BuildAntenna[type, numFinalParticles, numLoops]` already encodes the entire logical chain and is able to return all intermediate steps using the appropriate options.

First, by running the function naively we get the final result in Eq. 4.13,

```
In[2]:= BuildAntenna[A,3,0]//Collect[#,Epsilon]&
Out[2]:=
```

$$\frac{2s_{12}}{q^2 s_{13}}+\frac{2s_{12}}{q^2 s_{23}}+\frac{2s_{12}^2}{q^2 s_{13}s_{23}}+\frac{s_{13}}{q^2 s_{23}}+\frac{s_{23}}{q^2 s_{13}}+\epsilon\left(\frac{-2}{q^2}-\frac{s_{13}}{q^2 s_{23}}-\frac{s_{23}}{q^2 s_{13}}\right)$$

We can also use the `ReturnRecord` option to access all intermediate steps, and check that AntCalc goes through the same steps as the by-hand derivation in Sec. 4.2. We can see that with,

```
In[3]:=
BuildAntenna[A,3,0,ReturnRecord->True]["IntermediateSteps"]
Out[3]:=
  Amplitude:
```

$$T^{a_3}_{i_1 i_2}\left[\frac{\bar{u}(k_1)\,\not\epsilon^*(k_3)\,(\not k_1+\not k_3)\,\not\epsilon(p)\,v(k_2)}{(k_1+k_3)^2}-\frac{\bar{u}(k_1)\,\not\epsilon(p)\,(\not k_2+\not k_3)\,\not\epsilon^*(k_3)\,v(k_2)}{(k_2+k_3)^2}\right]$$

```
  Interference:
```

$$\frac{4(\epsilon-1)(N^2-1)\left(-2s_{12}^2-2s_{12}(s_{13}+s_{23})+(\epsilon-1)s_{13}^2+2\epsilon\, s_{13}s_{23}+(\epsilon-1)s_{23}^2\right)}{s_{13}s_{23}}$$

```
  Result:
```

$$-\frac{2\epsilon}{q^2}+\frac{2s_{12}^2}{q^2 s_{13}s_{23}}+\frac{2s_{12}}{q^2 s_{13}}+\frac{2s_{12}}{q^2 s_{23}}+\frac{s_{13}}{q^2 s_{23}}-\frac{\epsilon\, s_{13}}{q^2 s_{23}}+\frac{s_{23}}{q^2 s_{13}}-\frac{\epsilon\, s_{23}}{q^2 s_{13}}$$

where `Amplitude`, `Interference` and `Result` respectively refer to the expressions in Eqs. 4.1, 4.9 and 4.13.

4.3.2 Antenna Integration

Integrating antennae using AntCalc is also simple. Every step in 4.2.5 is handled by a single function: `IntegrateAntenna[antenna]`.

Before going over how this function is used, a small clarification needs to be made: the `IntegrateAntenna` function is not compatible with the default output of the `BuildAntenna` function described earlier. This is so that the built antenna is immediately useful and easily readable. As described in Chapter 3, however, the IBP reduction steps are handled by the LiteRed2 package, which does not expect Mandelstam invariants, but rather, momenta. Accordingly, to use `BuildAntenna`'s output in `IntegrateAntenna` we must toggle the relevant flag: `BuildAntenna[A,3,0,IntegrableForm->True]`. This will return a large association with all required information for integrating the antenna.

We may now proceed to the integration. The whole process goes as follows,

```
In[4]:= intA30 = BuildAntenna[A,3,0,IntegrableForm->True];
In[5]:= IntegrateAntenna[intA30,ExpansionOrder->2]
Out[5]:=
```

$$\frac{19}{4}+\epsilon^{-2}+\frac{3}{2\epsilon}-\frac{7\pi^2}{12}+\epsilon^2\left(\frac{639}{16}-\frac{133\pi^2}{48}-\frac{71\pi^4}{1440}-\frac{25\zeta_3}{2}\right)+\epsilon\left(\frac{109}{8}-\frac{7\pi^2}{8}-\frac{25\zeta_3}{3}\right)$$

Much like with `BuildAntenna`, `IntegrateAntenna` also allows us to inspect its intermediate steps. Here, however, the available options show the master integral linear combination expression both with R_3 and with it substituted by the appropriate expression, that is, the equivalent to Eqs. 4.16 and 4.18 before normalisation. The code and results are,

```
In[6]:= resA30 = IntegrateAntenna[intA30,ExpansionOrder->2,
ReturnRecord->True];
In[7]:= resA30["MasterCombination"]//Simplify//Collect[#,R3]&
Out[7]:=
```

$$\frac{-4(d-3)\big(4+d(\epsilon-2)-4\epsilon\big)}{(d-4)^2q^2}R_3$$

```
In[8]:= resA30["MasterSubstitutedExpression"]//Simplify
Out[8]:=
```

$$-\left(q^2\right)^{-2\epsilon}\frac{2^{-5+4\epsilon}(d-3)\big(4+d(\epsilon-2)-4\epsilon\big)\pi^{-3+2\epsilon}\Gamma^3(1-\epsilon)}{(d-4)^2\Gamma(3-3\epsilon)\Gamma(2-2\epsilon)}$$

The results here are given as a combination of d and ϵ. This is a result of the way the function processes each step. Since these are snapshots of what is happening inside the function itself, they show the expressions before dimension regularisation is applied.

Finally, AntCalc also enables us to forgo the work we did building and the integrating the antenna with the use of the `BuildAndIntegrateAntenna[type, numFinalParticles,numLoops]` function. It goes over the whole process, as we did, and outputs the ϵ Laurent series,

```
In[9]:= BuildAndIntegrateAntenna[A,3,0,ExpansionOrder->2]
Out[9]:=
```

$$\frac{19}{4}+\epsilon^{-2}+\frac{3}{2\epsilon}-\frac{7\pi^2}{12}+\epsilon^2\left(\frac{639}{16}-\frac{133\pi^2}{48}-\frac{71\pi^4}{1440}-\frac{25\zeta_3}{2}\right)+\epsilon\left(\frac{109}{8}-\frac{7\pi^2}{8}-\frac{25\zeta_3}{3}\right)$$

The agreement between the two sections in the present chapter confirms that AntCalc completes the same function we would do by-hand, going through the same steps. The same pipeline extends to the more complex antennae treated in this thesis, with their integrated results collected in Appendix B.

Chapter 5

Validation

The antenna function computed in the previous chapter and all others in SMQCD (and other models) find their primary application in infrared subtraction schemes for higher-order cross-section calculations. Real-emission contributions to fixed-order cross sections are individually divergent in the soft and collinear limits, and standard numerical integration fails at these phase-space boundaries. The subtraction approach resolves this by introducing a local counterterm with precisely the same singular behaviour as the real-emission matrix element at every phase-space point, so that their difference is finite and numerically integrable. For this cancellation to hold locally, the subtraction term must reproduce the universal QCD singular structure in each unresolved limit: the soft eikonal factor,

$$\mathcal{S}_{12}^{(3)} = 2\frac{s_{12}}{s_{13}s_{23}} \tag{5.1}$$

in the soft $k_3 \to 0$ limit, and the Altarelli-Parisi splitting function,

$$P_{qg}(z) = \frac{1+z^2}{1-z} \tag{5.2}$$

in the collinear $s_{13} \to 0$ limit. In the antenna subtraction formalism, the antenna functions represent these local counterterms. In Section 5.1 we verify that the unintegrated A_3^0 antenna generated by AntCalc reproduces both structures exactly.

Global consistency of the full antenna set is then tested via the computation of the massless hadronic R-ratio up to NNLO in Section 5.2. At each order the result is assembled from the integrated antenna functions computed using AntCalc and infrared poles must cancel exactly in the sum before a finite physical coefficient can be achieved.

5.1 Soft and Collinear Limits of A_3^0

Applying those limits to our A_3^0 unintegrated antenna enables us to verify those results.

First, the eikonal factor at the soft limit is computed as follows,

$$\mathcal{S}_{12}^{(3)} = \lim_{k_3\to 0} A_3^0(1_q, 3_g, 2_{\bar{q}}). \tag{5.3}$$

Using AntCalc, computing this eikonal factor goes as,

```
In[2]:= intA30 = BuildAntenna[A,3,0];

softSubs = s13 -> lambda * s13, s23 -> lambda * s23;
A30Soft = A30 /.  softSubs;

A30SoftSeries = Series[A30Soft, lambda, 0, 0];

eikonalFactor = SeriesCoefficient[A30SoftSeries, -2] /.  q2 -> s12
// Simplify;
Out[2]:=
```

$$\frac{2s_{12}}{s_{13}s_{23}}$$

Computing the Altarelli-Parisi splitting function is very similar do computing the eikonal factor, but using the collinear limit instead of the soft limit, as follows,

$$P_{qg}(z) = \lim_{k_1||k_3} A_3^0(1_q, 3_g, 2_{\bar{q}}) \tag{5.4}$$

and using AntCalc,

```
In[3]:= intA30 = BuildAntenna[A,3,0];

collinearSubs = s13 -> lambda, s12 -> z q2, s23 -> (1 - z) q2,
Epsilon -> 0;
A30Colliner = A30 /.  collinearSubs;

A30CollinearSeries = Series[A30Colliner, lambda, 0, 0];

apSplittingFunc = 1 / q2 SeriesCoefficient[A30CollinearSeries, -1]
// Simplify;
Out[3]:=
```

$$\frac{1+z^2}{1-z}$$

which both equal their respective known results in Eqs. 5.1 and 5.2.

5.2 The Massless SMQCD R-Ratio

The hadronic R-ratio is usually defined as the ratio of total cross-section for e^+e^- annihilation into hadrons to the point-like cross-section for $e^+e^- \longrightarrow \mu^+\mu^-$, that is, it is the ratio of the cross-sections of the topologies where the electron-positron pair annihilates into hadrons and into a muon-antimuon pair. It is mathematically described as follows,

$$\begin{aligned} R_{\text{hadr}} &= \frac{\sigma(e^+e^- \longrightarrow \text{hadrons})}{\sigma(e^+e^- \longrightarrow \mu^+\mu^-)} \\ &= \frac{3q^2}{4\pi\alpha_s^2}\sigma(e^+e^- \longrightarrow \text{hadrons}). \end{aligned} \tag{5.5}$$

Looking closely, the $\sigma(e^+e^- \to \text{hadrons})$ term is what QCD predicts for the $\gamma^* \to$ hadrons, computed via the optical theorem [ref to the optical theorem]. What makes the R-ratio a clean QCD observable, and a good benchmark for AntCalc is that the ratio features terms for higher perturbation orders as a perturbative series in α_s,

$$R_{\text{hadr}} = \left(c_{\text{LO}} + \frac{\alpha_s}{\pi} c_{\text{NLO}} + \left(\frac{\alpha_s}{\pi}\right)^2 c_{\text{NNLO}} + O\left(\alpha_s^3\right)\right) \tag{5.6}$$

here the perturbative coefficients, c_{LO}, c_{NLO} and c_{NNLO}, respectively receive contributions from the multiparticle final states at the relative orders, and respective antennae, as described in Chapter 2. For example, A_3^0 as described in Chapter 4 contributes to c_{NLO}. Moreover, the leading order term $R^{(0)}$ simply counts the number of quark colours and flavours weighted by their electric charges,

$$R^{(0)} = N_c \sum_q e_q^2 \tag{5.7}$$

from $R_{\text{hadr}} = R^{(0)}\left(1 + O(\alpha_s)\right) = R^{(0)} R(\alpha_s)$.

To compute the R-ratio, we will start by computing the three coefficients using the integrated antennae. Those integrated antennae will be assembled into $\mathcal{T}$-objects, as described in Chapter 2, and those will finally be used to assemble the R perturbative series.

In this chapter, as the title suggests we will be focusing on the SMQCD R-ratio, not SUSY or HiggsEFT. That means that we will only be using A-, B- and C-type antennae.

5.2.1 Computing the Integrated Antennae and $\mathcal{T}$-objects

As shown in Chapter 4, building and integrating antennae is made very easy using AntCalc. Since our goal is to use the all integrated antennae in SMQCD we can make use of the `BuildAndIntegrateAllAntennae[model]` function. The integrated antennae will be generated as follows,

```
In[2]:= {intA20,intA30,intA21,{intA40, intTildeA40, intHatA40},
intB40, intC40, {intA31, intTildeA31, intHatA31}, {intA22,
intTildeA22, intHatA22, intBreveA22}} =
BuildAndIntegrateAllAntennae[SMQCD, maxOrder->NNLO,
ExpansionOrder -> 2];
```

which generates all integrated antennae up to NNLO and up to the ϵ^2 term, in the SMQCD model.

The $\mathcal{T}$-objects, as described in Subsection 2.3.3, are then assembled as,

$$
\begin{aligned}
\mathcal{T}^2_{q\bar{q}} &= 4N(1-\epsilon) \\
\mathcal{T}^4_{q\bar{q}} &= \left(N - \frac{1}{N}\right) \mathcal{T}^2_{q\bar{q}} \mathcal{A}^1_2 \\
\mathcal{T}^4_{q\bar{q}g} &= \left(N - \frac{1}{N}\right) \mathcal{T}^2_{q\bar{q}} \mathcal{A}^0_3 \\
\mathcal{T}^6_{q\bar{q}} &= \left(N - \frac{1}{N}\right) \mathcal{T}^2_{q\bar{q}} \left[N\mathcal{A}^2_2 + \frac{1}{N}\tilde{\mathcal{A}}^2_2 + N_f \hat{\mathcal{A}}^2_2 + \left(N - \frac{1}{N}\right) \breve{\mathcal{A}}^2_2 \right] \\
\mathcal{T}^6_{q\bar{q}g} &= \left(N - \frac{1}{N}\right) \mathcal{T}^2_{q\bar{q}} \left[N \left(\mathcal{A}^1_3 + \mathcal{A}^1_2 \mathcal{A}^0_3\right) - \frac{1}{N} \left(\tilde{\mathcal{A}}^1_3 + \mathcal{A}^1_2 \mathcal{A}^0_3\right) + N_f \hat{\mathcal{A}}^1_3 \right] \\
\mathcal{T}^6_{q\bar{q}q'\bar{q}'} &= \left(N - \frac{1}{N}\right) \mathcal{T}^2_{q\bar{q}} (N_f - 1) \mathcal{B}^0_4 \\
\mathcal{T}^6_{q\bar{q}q\bar{q}} &= \frac{1}{N_f - 1} \mathcal{T}^6_{q\bar{q}q\bar{q}'q'} - \mathcal{T}^2_{q\bar{q}} \left(N - \frac{1}{N}\right) \frac{1}{N} \mathcal{C}^0_4 \\
\mathcal{T}^6_{q\bar{q}gg} &= \left(N - \frac{1}{N}\right) \mathcal{T}^2_{q\bar{q}} \left(N\mathcal{A}^0_4 - \frac{1}{N}\tilde{\mathcal{A}}^0_4\right)
\end{aligned}
\tag{5.8}
$$

and in terms of the *Wolfram* language,

```
In[3]:=
Tqq2 = 4 n * (1 - eps);
Tqq4 = (n - 1 / n) Tqq2 intA21;
Tqqg4 = (n - 1 / n) Tqq2 intA30;
Tqq6 = (n - 1 / n) Tqq2 * (n * intA22 + 1 / n * intTildeA22 + nf *
intHatA22) + Tqq2 * (n - 1 / n) ^ 2 * intBreveA22;
Tqqg6 = (n - 1 / n) Tqq2 (n * (intA31 + intA21 intA30) - 1 / n *
(intTildeA31 + intA21 intA30) + nf intHatA31);
Tqqqqprime6 = (n - 1 / n) Tqq2 * (nf- 1) * intB40;
Tqqqq6 = 1 / (nf - 1) * Tqqqqprime6 - Tqq2 (n - 1 / n) * 1 / n *
intC40;
Tqqgg6 = (n - 1 / n) Tqq2 * (n intA40 - 1 / n *
intTildeA40);
```

5.2.2 Assembling the R-ratio

With the $\mathcal{T}$-objects in hand, we now have everything we need to generate the R-ratio for SMQCD. To finally do that, we use the following formula,

$$
\begin{aligned}
R = \frac{R_{\text{hadr}}}{R^{(0)}} &= R_{\text{LO}} + R_{\text{NLO}} + R_{\text{NNLO}} \\
&= r_{\text{LO}} + \frac{\alpha_s}{2\pi} r_{\text{NLO}} + \left(\frac{\alpha_s}{2\pi}\right)^2 r_{\text{NNLO}} \\
&= 1 + \frac{\alpha_s}{2\pi} \frac{\mathcal{T}^4_{q\bar{q}} + \mathcal{T}^4_{q\bar{q}g}}{\mathcal{T}^2_{q\bar{q}}} + \left(\frac{\alpha_s}{2\pi}\right)^2 \frac{\mathcal{T}^6_{q\bar{q}} + \mathcal{T}^6_{q\bar{q}g} + \mathcal{T}^6_{q\bar{q}q\bar{q}} + \mathcal{T}^6_{q\bar{q}q'\bar{q}'} + \mathcal{T}^6_{q\bar{q}gg}}{\mathcal{T}^2_{q\bar{q}}} + \mathcal{O}\left(\alpha_s^3\right).
\end{aligned}
\tag{5.9}
$$

Substituting the integrated antennae in we are able to show that the perturbative orders equal,

$$\begin{aligned} r_{\mathrm{LO}} &= 1 \\ r_{\mathrm{NLO}} &= \left(N - \frac{1}{N}\right)\left(\mathcal{A}_2^1 + \mathcal{A}_3^0\right) \\ r_{\mathrm{NNLO}} &= \left(N - \frac{1}{N}\right)\Bigg[N\left(\mathcal{A}_2^1\mathcal{A}_3^0 + \mathcal{A}_3^1 + \mathcal{A}_4^0 + \mathcal{A}_2^2\right) - \frac{1}{N}\left(\mathcal{A}_2^1\mathcal{A}_3^0 + \tilde{\mathcal{A}}_3^1 + \tilde{\mathcal{A}}_4^0 + C_4^0 - \tilde{\mathcal{A}}_2^2\right) \\ &\qquad + N_f\left(\hat{\mathcal{A}}_3^1 + \hat{\mathcal{A}}_2^2\right) + \left(N - \frac{1}{N}\right)\breve{\mathcal{A}}_2^2\Bigg] \end{aligned} \tag{5.10}$$

these coefficients are then used to finally assemble the R-ratio, where the poles cancel to make a finite result, as follows

$$R = 1 + \frac{\alpha_s}{2\pi}\left(\frac{N^2-1}{2N}\right)\frac{3}{2} + \left(\frac{\alpha_s}{2\pi}\right)^2\left(\frac{N^2-1}{2N}\right)\left[N\left(\frac{243}{16} - 11\zeta_3\right) + \frac{3}{16N} + N_f\left(-\frac{11}{4} + 2\zeta_3\right)\right] \tag{5.11}$$

and in *Wolfram*,

```
In[4]:=
cLO = 1;
rLO = 1;

cNLO = alphaS / 2 Pi;
rNLO = (Tqq4 + Tqqg4) / Tqq2;

cNNLO = (cNLO) ^ 2;
rNNLO = (Tqq6 + Tqqg6 + Tqqqq6 + Tqqqqprime6 + Tqqgg6) / Tqq2;

R = cLO rLO + cNLO rNLO + cNNLO rNNLO//Collect[#,alphaS / 2 Pi]&

Out[4]:=
```

$$1 + \frac{\alpha_s}{2\pi}\left(\frac{N^2-1}{2N}\right)\frac{3}{2} + \left(\frac{\alpha_s}{2\pi}\right)^2\left(\frac{N^2-1}{2N}\right)\left[N\left(\frac{243}{16} - 11\zeta_3\right) + \frac{3}{16N} + N_f\left(-\frac{11}{4} + 2\zeta_3\right)\right]$$

Using a Shortcut

As shown before in this text, AntCalc often includes quality of life features that make the work we are taking many steps doing into a single all-inclusive function. The R-ratio calculation is no different.

First, AntCalc offers the ability to compute the $\mathcal{T}$-objects directly using the `TObject[order, finalParticles]`. That is applied as follows,

```
In[5]:=
Tqq2 = TObject[2, q, qbar, ExpansionOrder -> 2];
Tqq4 = TObject[4, q, qbar, ExpansionOrder -> 2];
Tqqg4 = TObject[4, q, qbar, g, ExpansionOrder -> 2];
```

```
Tqq6 = TObject[6, q, qbar, ExpansionOrder -> 2];
Tqqg6 = TObject[6, q, qbar, g, ExpansionOrder -> 2];
Tqqqqprime6 = TObject[6, q, qbar, qprime, qbarprime,
ExpansionOrder -> 2];
Tqqqq6 = TObject[6, q, qbar, q, qbar, ExpansionOrder -> 2];
Tqqgg6 = TObject[6, q, qbar, g, g, ExpansionOrder -> 2];
```

noting here that the default `ExpansionOrder` flag is 0.

Moreover, the function `BuildRRatio[model]` does exactly what we went through in the present chapter and yields the same result. Its use goes as follows,

```
In[6]:= BuildRRatio[SMQCD]

Out[6]:=
```

$$1 + \frac{3\left(N^2-1\right)\alpha_s^2}{8N\pi} + \frac{\left(N^2-1\right)\alpha_s^2}{8N\pi}\left[\frac{3}{16N} + N\left(\frac{243}{16} - 11\zeta_3\right) + N_f\left(-\frac{11}{4} + 2\zeta_3\right)\right]$$

Chapter 6

Future Work

AntCalc, as described in Chapter 3 and validated in Chapter 5, for both its the building and integration engines, currently encompasses all massless antennae in SMQCD up to NNLO.

Modern fixed-order QCD calculations, however, increasingly require antenna functions beyond this scope. Massive quark effects enter heavy-flavour production at hadron colliders, supersymmetric models introduce new parton species, and Higgs-plus-jet processes at NNLO require antennae derived from an effective $H \to gg$ vertex. Extension to N^{3}LO is also an active frontier, as the precision demands of the LHC physics programme continue to tighten. The natural directions for extending AntCalc are therefore the massive SMQCD sector, the massless SUSY and HiggsEFT models, and eventually the full N^{3}LO antenna set.

The physical, mathematical and computation challenges introduced by these extensions are discussed to a higher level of detail in Appendix D.

6.1 Massive Antennae

As discussed in Chapter 4, the A_3^0 antenna is the simplest non-trivial SMQCD antenna in the model, and as such, is also the best topology to use to extend AntCalc to the massive framework. This extension has already been implemented in the current version. To access the results, we can run the usual functions, using the `quarkMass` option,

```
In[2]:= BuildAntenna[A,3,0,quarkMass->mQ]
Out[2]:=
```

$$\frac{1}{4s_{13}^2 s_{23}^2\left(4m_q^2+s_{12}+s_{13}+s_{23}\right)}\Big[-8m_q^2\left(s_{13}^2+s_{23}^2\right)+s_{13}s_{23}\left(2s_{12}^2+s_{13}^2+s_{23}^2+2s_{12}(s_{13}+s_{23})\right)$$
$$-2m_q^2\left(s_{13}^3+s_{13}^2s_{23}+s_{13}s_{23}^2+s_{23}^3+s_{12}\left(s_{13}^2-4s_{13}s_{23}+s_{23}^2\right)\right)\Big]$$

```
In[3]:=
BuildAndIntegrateAntenna[A,3,0,quarkMass->mQ]
/.{j[MX30Basis123, 1, 1, 1, 0, 0]->RM1,
j[MX30Basis123, 2, 1, 1, 0, 0]->RM2,d->4-2Epsilon, eps->Epsilon}
//FullSimplify//Collect[#,{RM1,RM2}]&
```

```
Out[3]:=
```

$$\frac{8(2-\epsilon(7-6\epsilon))m_q^4-4\Big(4+\epsilon\big(-11+4\epsilon(1+\epsilon)\big)\Big)m_q^2q^2-(1-2\epsilon)\big(4-\epsilon(3+2\epsilon)\big)q^4}{\epsilon(1+2\epsilon)\left(4m_q^2-q^2\right)q^4}R_1^M$$
$$+\frac{16(1-2\epsilon)m_q^6+8\big(-1+4\epsilon(1+\epsilon)\big)m_Q^4q^2-4\left(1+2\epsilon+4\epsilon^2\right)m_q^2q^4+2q^6+\epsilon(-1+2\epsilon)q^6}{\epsilon(1-2\epsilon)\left(4m_q^2-q^2\right)q^4}R_2^M$$

To get to these results, the same method from Chapter 4 was used, but for the massive case, in which $k^2 \neq 0 \neq m_q$. The results still are not complete, as the massive master integrals, R_1^M and R_2^M, have yet to be computed to enable us to expand into the usual ϵ Laurent series. This example does show, however, that the already present framework is able to handle the massive expansion, at least inside SMQCD, seamlessly, opening it up to the possibility of further expansion to the massive SMQCD (and beyond) sector.

6.2 SUSY model

6.2.1 Short Introduction To SUSY

6.2.2 Example Implementation

6.3 HiggsEFT model

6.3.1 Short Introduction To HiggsEFT

6.3.2 Example Implementation

6.4 N^{3}LO

Chapter 7

Conclusion

Conclusion goes here.

Appendix A

Master Integral Derivations

Appendix A goes here.

Appendix B

Integrated Antenna Trees

Appendix B goes here.

Appendix C

IBP Derivation

Appendix C goes here.

Appendix D

Future Work Details

Appendix D goes here.

Bibliography

[1] R. Keith Ellis, W. James Stirling, and B. R. Webber. *QCD and collider physics*. Vol. 8. Cambridge University Press, Feb. 2011. ISBN: 978-0-511-82328-2, 978-0-521-54589-1. DOI: 10.1017/CBO9780511628788.

[2] Matthew D. Schwartz. *Quantum Field Theory and the Standard Model*. Cambridge University Press, Mar. 2014. ISBN: 978-1-107-03473-0, 978-1-107-03473-0. DOI: 10.1017/9781139540940.

[3] Michael E. Peskin and Daniel V. Schroeder. *An Introduction to quantum field theory*. Reading, USA: Addison-Wesley, 1995. ISBN: 978-0-201-50397-5, 978-0-429-50355-9, 978-0-429-49417-8. DOI: 10.1201/9780429503559.

[4] Gudrun Heinrich. "Collider physics at the precision frontier". In: *Physics Reports* 922 (Aug. 2021), pp. 1–69. ISSN: 0370-1573. DOI: 10.1016/j.physrep.2021.03.006. URL: http://dx.doi.org/10.1016/j.physrep.2021.03.006.

[5] A. and Abada et al. "FCC-ee: The Lepton Collider: Future Circular Collider Conceptual Design Report Volume 2". In: *The European Physical Journal Special Topics* 228.2 (2019), pp. 261–623. ISSN: 1951-6401. DOI: 10.1140/epjst/e2019-900045-4. URL: http://dx.doi.org/10.1140/epjst/e2019-900045-4.

[6] Toichiro Kinoshita. "Mass Singularities of Feynman Amplitudes". In: *Journal of Mathematical Physics* 3.4 (1962), pp. 650–677. ISSN: 1089-7658. DOI: 10.1063/1.1724268. URL: http://dx.doi.org/10.1063/1.1724268.

[7] T. D. Lee and M. Nauenberg. "Degenerate Systems and Mass Singularities". In: *Phys. Rev.* 133 (1964). Ed. by G. Feinberg, B1549–B1562. DOI: 10.1103/PhysRev.133.B1549.

[8] S. Catani and M. H. Seymour. "A General algorithm for calculating jet cross-sections in NLO QCD". In: *Nucl. Phys. B* 485 (1997). [Erratum: Nucl.Phys.B 510, 503–504 (1998)], pp. 291–419. DOI: 10.1016/S0550-3213(96)00589-5. arXiv: hep-ph/9605323.

[9] Gerard 't Hooft and M. J. G. Veltman. "Regularization and Renormalization of Gauge Fields". In: *Nucl. Phys. B* 44 (1972), pp. 189–213. DOI: 10.1016/0550-3213(72)90279-9.

[10] Zvi Bern and David A. Kosower. "The Computation of loop amplitudes in gauge theory". In: *Nucl. Phys. B* 379 (1992), pp. 451–461. DOI: 10.1016/0550-3213(92)90240-Y.

[11] A. Gehrmann-De Ridder, T. Gehrmann, and E. W. Nigel Glover. "Antenna subtraction at NNLO". In: *JHEP* 09 (2005), p. 056. DOI: 10.1088/1126-6708/2005/09/056. arXiv: hep-ph/0505111.

[12] William A. Bardeen et al. "Deep Inelastic Scattering Beyond the Leading Order in Asymptotically Free Gauge Theories". In: *Phys. Rev. D* 18 (1978), pp. 3998–4019. DOI: 10.1103/PhysRevD.18.3998.

[13] David A. Kosower. “Antenna factorization of gauge theory amplitudes”. In: *Phys. Rev. D* 57 (1998), pp. 5410–5416. DOI: 10.1103/PhysRevD.57.5410. arXiv: hep-ph/9710213.

[14] John M. Campbell, M. A. Cullen, and E. W. N. Glover. “Four jet event shapes in electron - positron annihilation”. In: *Eur. Phys. J. C* 9 (1999), pp. 245–265. DOI: 10.1007/s100520050630. arXiv: hep-ph/9809429.

[15] John Clive Ward. “An Identity in Quantum Electrodynamics”. In: *Phys. Rev.* 78 (1950), p. 182. DOI: 10.1103/PhysRev.78.182.

[16] Yasushi Takahashi. “On the generalized Ward identity”. In: *Nuovo Cim.* 6 (1957), pp. 371–375. DOI: 10.1007/BF02832514.

[17] J. C. Taylor. “Ward Identities and Charge Renormalization of the Yang-Mills Field”. In: *Nucl. Phys. B* 33 (1971), pp. 436–444. DOI: 10.1016/0550-3213(71)90297-5.

[18] A. A. Slavnov. “Ward Identities in Gauge Theories”. In: *Theor. Math. Phys.* 10 (1972), pp. 99–104. DOI: 10.1007/BF01090719.

[19] A. Gehrmann-De Ridder, T. Gehrmann, and E.W.N. Glover. “Quark–gluon antenna functions from neutralino decay”. In: *Physics Letters B* 612.1-2 (Apr. 2005), pp. 36–48. ISSN: 0370-2693. DOI: 10.1016/j.physletb.2005.02.039. URL: http://dx.doi.org/10.1016/j.physletb.2005.02.039.

[20] Federico Demartin et al. “Higgs characterisation at NLO in QCD: CP properties of the top-quark Yukawa interaction”. In: *The European Physical Journal C* 74.9 (2014). ISSN: 1434-6052. DOI: 10.1140/epjc/s10052-014-3065-2. URL: http://dx.doi.org/10.1140/epjc/s10052-014-3065-2.

[21] E. W. Nigel Glover and Joao Pires. “Antenna subtraction for gluon scattering at NNLO”. In: *Journal of High Energy Physics* 2010.6 (2010). ISSN: 1029-8479. DOI: 10.1007/jhep06(2010)096. URL: http://dx.doi.org/10.1007/jhep06(2010)096.

[22] A. Gehrmann-De Ridder, T. Gehrmann, and E. W. Nigel Glover. “Infrared structure of e+ e- —> 2 jets at NNLO”. In: *Nucl. Phys. B* 691 (2004), pp. 195–222. DOI: 10.1016/j.nuclphysb.2004.05.017. arXiv: hep-ph/0403057.

[23] A. Gehrmann-De Ridder, T. Gehrmann, and G. Heinrich. “Four-Particle Phase Space Integrals in Massless QCD”. In: (2003). DOI: 10.48550/ARXIV.HEP-PH/0311276. URL: https://arxiv.org/abs/hep-ph/0311276.

[24] T. Gehrmann, T. Huber, and D. Maitre. “Two-Loop Quark and Gluon Form Factors in Dimensional Regularisation”. In: (2005). DOI: 10.48550/ARXIV.HEP-PH/0507061. URL: https://arxiv.org/abs/hep-ph/0507061.

[25] G. Passarino and M. J. G. Veltman. “One-Loop Corrections for e^+e^- Annihilation into $\mu^+\mu^-$ in the Weinberg Model”. In: *Nucl. Phys. B* 160 (1979), pp. 151–207. DOI: 10.1016/0550-3213(79)90234-7.

[26] K. G. Chetyrkin and F. V. Tkachov. “Integration by Parts: The Algorithm to Calculate β-Functions in 4 Loops”. In: *Nucl. Phys. B* 192 (1981), pp. 159–204. DOI: 10.1016/0550-3213(81)90199-1.

[27] Charalampos Anastasiou and Kirill Melnikov. “Higgs Boson Production at Hadron Colliders in NNLO QCD”. In: *Nucl. Phys. B* 646 (2002), pp. 220–256. DOI: 10.1016/S0550-3213(02)00837-4. arXiv: hep-ph/0207004.

[28] Stefano Laporta. "High-Precision Calculation of Multi-Loop Feynman Integrals by Difference Equations". In: *Int. J. Mod. Phys. A* 15 (2000), pp. 5087–5159. DOI: 10.1142/S0217751X00002159. arXiv: hep-ph/0102033.

[29] Ettore Remiddi. "Differential Equations for Feynman Graph Amplitudes". In: *Nuovo Cim. A* 110 (1997), pp. 1435–1452. DOI: 10.1007/BF03185566. arXiv: hep-th/9711188.

[30] A. Gehrmann-De Ridder and M. Ritzmann. "NLO Antenna Subtraction with Massive Fermions". In: *JHEP* 07 (2009), p. 041. DOI: 10.1088/1126-6708/2009/07/041. arXiv: 0904.3297 [hep-ph].